

TYPE	BANK	B23A Package	B32A Package
Transceiver I/O	1A	-	20
Transceiver I/O	1B	20	20
Transceiver I/O	1C	22	22
HSIO	2A_B	-	48
HSIO	2A_T	-	48
HSIO	2B_B	-	48
HSIO	2B_T	-	48
HPS shared HSIO	3A_B	48	48
HPS and SDM shared HSIO	3A_T	48	48
HPS shared HSIO	3B_B	-	48
HPS shared HSIO	3B_T	-	48
Transceiver I/O	4A	-	20
Transceiver I/O	4B	-	22
Transceiver I/O	4C	22	20
HVIO	5A	20	20
HVIO	5B	20	20
HVIO	6A	20	20
HVIO	6B	20	20
HVIO	6C	20	20
HVIO	6D	20	20
HPS I/O	HPS	48	48
SDM I/O	SDM	33	33

- i. Total LVDS channels per bank supporting LVDS SERDES Non-DPA and DPA mode is equivalent to (LVDS I/O per bank)/2, inclusive of clock pair. Please refer to Dedicated Tx/Rx Channel column in the pin-out table for the channel availability.
- ii. Total LVDS channels supporting LVDS SERDES Soft-CDR mode is 12 pairs per bank. Please refer to Soft CDR column in the pin out table for the channel availability.

altera Pin Information for Agilex™ 5 A5ED052A Device Version: 2025-06-06 B23A Status: Final												
Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B23A	DQS for X4	DQS for X8/X9	DQS for X16/X18	
SDM		TDO						BN55				
SDM		TMS						BJ58				
SDM		TCK						BJ58				
SDM		TDI						BJ53				
SDM		OSC_CLK_1						BJ48				
SDM		SDM_IO1	AVSTX8_DATA2,AS_DATA1					BV29				
SDM		SDM_IO5	AS_nCS00,MSEL0					BJ30				
SDM		SDM_IO3	AVSTX8_DATA3,AS_DATA2					BR36				
SDM		nCONFIG						BU32				
SDM		SDM_IO4	AVSTX8_DATA1,AS_DATA0					BN43				
SDM		SDM_IO2	AVSTX8_DATA0,AS_CLK					BV26				
SDM		SDM_IO7	AS_nCS02,MSEL1					BN31				
SDM		SDM_IO11	AVSTX8_VALIO,PWRMGMT_SDA					BN36				
SDM		nSTATUS						BV32				
SDM		SDM_IO16	PWRMGMT_SDA					BN46				
SDM		SDM_IO0	PWRMGMT_SCL,PWRMGMT_ALERT					BU34				
SDM		SDM_IO13	AVSTX8_DATA5					BJ45				
SDM		SDM_IO9	AS_nCS01,MSEL2,PWRMGMT_ALERT					BU29				
SDM		SDM_IO6	AVSTX8_DATA4,AS_DATA3					BN38				
SDM		SDM_IO10	AVSTX8_DATA7					BN28				
SDM		SDM_IO8	AVSTX8_READY,AS_nCS03					BJ39				
SDM		SDM_IO12	PWRMGMT_SDA,PWRMGMT_ALERT					BJ33				
SDM		SDM_IO15	AVSTX8_DATA6,AS_nRST					BJ41				
SDM		SDM_IO14	AVSTX8_CLK,PWRMGMT_SCL					BR28				
SDM		RREF_SDM						BV51				
SDM			TEMPODIODEDAn					BR52				
SDM			TEMPODIODEAp					BN52				
SDM			VREFP_ADC					BU42				
SDM			VREFN_ADC					BU44				
SDM		VSIGP_0						BU47				
SDM		VSIGN_0						BV47				
SDM		VSIGP_1						BV42				
SDM		VSIGN_1						BV40				
HPS		HPS_IOA_1	GPIO0_IO0,SPIM0_SS1_N,SPIS0_CLK,UART0_CTS_N,NAND_ADQ0,SDMMC_DATA0,USB0_CLK,EMAC0_PP50,TRACE_D10		HPS_IOA_1			B80				
HPS		HPS_IOA_2	GPIO0_IO1,SPIM1_SS1_N,SPIS0_MISO,UART0_RTS_N,NAND_ADQ1,SDMMC_DATA1,USB0_STP,EMAC0_PPSTRIG0,TRACE_D9		HPS_IOA_2			C80				
HPS		HPS_IOA_3	GPIO0_IO2,SPIS0_SS0_N,UART0_TX,I2C1_SDA,NAND_WE_N,SDMMC_CLK,USB0_DIR,EMAC1_PPS1,TRACE_D8		HPS_IOA_3			E81				
HPS		HPS_IOA_4	GPIO0_IO3,SPIS0_MISO,UART0_RX,I2C1_SCL,NAND_RE_N,USB0_DATA0,EMAC1_PPSTRIG1,TRACE_D7		HPS_IOA_4			V64				
HPS		HPS_IOA_5	GPIO0_IO4,SPIM0_CLK,UART1_CTS_N,I2C0_SDA,NAND_WP_N,SDMMC_WRITE_PROTECT,USB0_DATA1,EMAC2_PPS2,TRACE_D6		HPS_IOA_5			G81				
HPS		HPS_IOA_6	GPIO0_IO5,SPIM0_MISO,UART1_RTS_N,I2C0_SCL,NAND_ADQ2,SDMMC_DATA2,USB0_NXT,EMAC2_PPSTRIG2,TRACE_D5		HPS_IOA_6			F64				
HPS		HPS_IOA_7	GPIO0_IO6,SPIM0_MISO,MDO10_MDO10,UART1_TX,I2C_EMAC2_SDA,NAND_ADQ3,SDMMC_DATA3,USB0_DATA2,TRACE_D4		HPS_IOA_7			G80				
HPS		HPS_IOA_8	GPIO0_IO7,SPIM0_SS0_N,MDO10_MDC,UART1_RX,I2C_EMAC2_SCL,NAND_CLE,SDMMC_CMD,USB0_DATA3,TRACE_D15		HPS_IOA_8			K74				
HPS		HPS_IOA_9	GPIO0_IO8,SPIM1_CLK,SPIS1_CLK,MDO10_MDO10,I2C_EMAC1_SDA,NAND_ADQ4,SDMMC_DATA4,USB0_DATA4,I3C1_SDA,TRACE_D14		HPS_IOA_9			E80				
HPS		HPS_IOA_10	GPIO0_IO9,SPIM1_MISO,SPIS1_MISO,MDO10_MDC,I2C_EMAC1_SCL,NAND_ADQ5,SDMMC_DATA5,USB0_DATA5,I3C1_SCL,TRACE_D13		HPS_IOA_10			F74				
HPS		HPS_IOA_11	GPIO0_IO10,SPIM1_MISO,SPIS1_SS0_N,MDO10_MDO10,I2C_EMAC0_SDA,NAND_ADQ6,SDMMC_DATA6,USB0_DATA6,I3C0_SDA,TRACE_D12		HPS_IOA_11			H78				
HPS		HPS_IOA_12	GPIO0_IO11,SPIM1_SS0_N,SPIS1_MISO,MDO10_MDC,I2C_EMAC0_SCL,NAND_ADQ7,SDMMC_DATA7,USB0_DATA7,I3C0_SCL,TRACE_D11		HPS_IOA_12			F78				
HPS		HPS_IOA_13	GPIO0_IO12,NAND_ALE,SDMMC_PU_PD_DATA2,USB1_CLK,EMAC0_TX_CLK,TRACE_D10		HPS_IOA_13			D74				
HPS		HPS_IOA_14	GPIO0_IO13,NAND_RB_N,SDMMC_PWR_ENA,USB1_STP,EMAC0_TX_CTL,TRACE_D9		HPS_IOA_14			B78				
HPS		HPS_IOA_15	GPIO0_IO14,NAND_CE_N,USB1_DIR,EMAC0_RX_CLK,TRACE_D8		HPS_IOA_15			B76				
HPS		HPS_IOA_16	GPIO0_IO15,NAND_DQ5,SDMMC_DATA_STROBE,USB1_DATA0,EMAC0_RX_CTL,TRACE_D7		HPS_IOA_16			K70				
HPS		HPS_IOA_17	GPIO0_IO16,I3C1_SDA,NAND_ADQ8,USB1_DATA1,EMAC0_TXD0,TRACE_D6		HPS_IOA_17			A76				
HPS		HPS_IOA_18	GPIO0_IO17,I3C1_SCL,NAND_ADQ9,USB1_NXT,EMAC0_TXD1,TRACE_D5		HPS_IOA_18			H70				
HPS		HPS_IOA_19	GPIO0_IO18,I3C0_SDA,NAND_ADQ10,USB1_DATA2,EMAC0_RXD0,TRACE_D4		HPS_IOA_19			B73				
HPS		HPS_IOA_20	GPIO0_IO19,SPIM1_SS1_N,I3C0_SCL,NAND_ADQ11,USB1_DATA3,EMAC0_RXD1,TRACE_CLK		HPS_IOA_20			K60				
HPS		HPS_IOA_21	GPIO0_IO20,SPIM1_CLK,SPIS0_CLK,UART0_CTS_N,I2C1_SDA,NAND_ADQ12,USB1_DATA4,EMAC0_RXD2,TRACE_D0		HPS_IOA_21			A73				
HPS		HPS_IOA_22	GPIO0_IO21,SPIM1_MISO,SPIS0_MISO,UART0_RTS_N,I2C1_SCL,NAND_ADQ13,USB1_DATA5,EMAC0_RXD3,TRACE_D1		HPS_IOA_22			F62				
HPS		HPS_IOA_23	GPIO0_IO22,SPIM1_MISO,SPIS0_SS0_N,UART0_TX,I2C0_SDA,NAND_ADQ14,USB1_DATA6,EMAC0_RXD2,TRACE_D2		HPS_IOA_23			B71				
HPS		HPS_IOA_24	GPIO0_IO23,SPIM1_SS0_N,SPIS0_MISO,UART0_RX,I2C0_SCL,NAND_ADQ15,USB1_DATA7,EMAC0_RXD3,TRACE_D3		HPS_IOA_24			A71				
HPS		HPS_IOB_1	GPIO1_IO0,SPIM1_CLK,UART0_CTS_N,EMAC0_PP50,NAND_ADQ0,SDMMC_DATA0,I3C0_SDA_PULLUP_EN,EMAC1_TX_CLK,TRACE_D10		HPS_IOB_1			B68				
HPS		HPS_IOB_2	GPIO1_IO1,SPIM1_MISO,UART0_RTS_N,EMAC0_PPSTRIG0,NAND_ADQ1,SDMMC_DATA1,I3C1_SDA_PULLUP_EN,EMAC1_TX_CTL,TRACE_D9		HPS_IOB_2			A63				
HPS		HPS_IOB_3	GPIO1_IO2,SPIM1_MISO,UART0_TX,I2C0_SDA,NAND_WE_N,SDMMC_CLK,EMAC1_RX_CLK,TRACE_D8		HPS_IOB_3			A51				
HPS		HPS_IOB_4	GPIO1_IO3,SPIM1_SS0_N,UART0_RX,I2C0_SCL,NAND_RE_N,EMAC1_RX_CTL,TRACE_D7		HPS_IOB_4			K67				
HPS		HPS_IOB_5	GPIO1_IO4,SPIM1_SS1_N,SPIS1_CLK,UART1_CTS_N,EMAC2_PPS2,NAND_WP_N,SDMMC_WRITE_PROTECT,I3C1_SDA,EMAC1_TXD0,TRACE_D6		HPS_IOB_5			B57				
HPS		HPS_IOB_6	GPIO1_IO5,SPIS1_MISO,UART1_RTS_N,EMAC2_PPSTRIG2,NAND_ADQ2,SDMMC_DATA2,I3C1_SCL,EMAC1_TXD1,TRACE_D5		HPS_IOB_6			F70				
HPS		HPS_IOB_7	GPIO1_IO6,SPIS1_SS0_N,UART1_TX,I2C1_SDA,NAND_ADQ3,SDMMC_DATA3,I3C0_SDA,EMAC1_RXD0,TRACE_D4		HPS_IOB_7			B65				
HPS		HPS_IOB_8	GPIO1_IO7,SPIS1_MISO,UART1_RX,I2C1_SCL,NAND_CLE,SDMMC_CMD,I3C0_SCL,EMAC1_RXD1,TRACE_D15		HPS_IOB_8			K62				
HPS		HPS_IOB_9	GPIO1_IO8,JTAG_TCK,SPIS0_CLK,MDO10_MDO10,I2C_EMAC2_SDA,NAND_ADQ4,SDMMC_DATA4,EMAC1_TXD2,TRACE_D14		HPS_IOB_9			A65				
HPS		HPS_IOB_10	GPIO1_IO9,JTAG_TMS,SPIS0_MISO,MDO10_MDC,I2C_EMAC2_SCL,NAND_ADQ5,SDMMC_DATA5,EMAC1_TXD3,TRACE_D13		HPS_IOB_10			H62				
HPS		HPS_IOB_11	GPIO1_IO10,JTAG_TDO,SPIS0_SS0_N,MDO10_MDO10,I2C_EMAC0_SDA,NAND_ADQ6,SDMMC_DATA6,EMAC1_RXD2,TRACE_D12		HPS_IOB_11			B61				
HPS		HPS_IOB_12	GPIO1_IO11,JTAG_TDI,SPIS0_MISO,MDO10_MDC,I2C_EMAC0_SCL,NAND_ADQ7,SDMMC_DATA7,EMAC1_RXD3,TRACE_D11		HPS_IOB_12			A61				
HPS		HPS_IOB_13	GPIO1_IO12,I2C1_SDA,NAND_ALE,SDMMC_PU_PD_DATA2,EMAC2_TX_CLK,TRACE_D10		HPS_IOB_13			B59				
HPS		HPS_IOB_14	GPIO1_IO13,I2C1_SCL,NAND_RB_N,SDMMC_PWR_ENA,EMAC2_TX_CTL,TRACE_D9		HPS_IOB_14			A59				
HPS		HPS_IOB_15	GPIO1_IO14,UART1_TX,NAND_CE_N,I3C1_SDA,EMAC2_RX_CLK,TRACE_D8		HPS_IOB_15			A54				
HPS		HPS_IOB_16	GPIO1_IO15,UART1_RX,NAND_DQ5,SDMMC_DATA_STROBE,I3C1_SCL,EMAC2_RX_CTL,TRACE_D7		HPS_IOB_16			F60				
HPS		HPS_IOB_17	GPIO1_IO16,UART1_CTS_N,NAND_ADQ8,I3C0_SDA,EMAC2_TXD0,TRACE_D6		HPS_IOB_17			B54				
HPS		HPS_IOB_18	GPIO1_IO17,SPIM0_SS1_N,UART1_RTS_N,NAND_ADQ9,I3C0_SCL,EMAC2_TXD1,TRACE_D5		HPS_IOB_18			D60				
HPS		HPS_IOB_19	GPIO1_IO18,SPIM0_MISO,MDO10_MDO10,I2C_EMAC1_SDA,NAND_ADQ10,EMAC2_RXD0,TRACE_D4		HPS_IOB_19			A49				
HPS		HPS_IOB_20	GPIO1_IO19,SPIM0_SS0_N,MDO10_MDC,I2C_EMAC1_SCL,NAND_ADQ11,EMAC2_RXD1,TRACE_CLK		HPS_IOB_20			F67				
HPS		HPS_IOB_21	GPIO1_IO20,SPIM0_CLK,SPIS1_CLK,I2C_EMAC2_SDA,NAND_ADQ12,I3C0_SDA_PULLUP_EN,EMAC2_TXD2,TRACE_D0		HPS_IOB_21			B49				
HPS		HPS_IOB_22	GPIO1_IO21,SPIM0_MISO,SPIS1_MISO,I2C_EMAC2_SCL,NAND_ADQ13,I3C1_SDA_PULLUP_EN,EMAC2_TXD3,TRACE_D1		HPS_IOB_22			D67				
HPS		HPS_IOB_23	GPIO1_IO22,SPIM0_MISO,SPIS1_SS0_N,MDO10_MDO10,I2C_EMAC0_SDA,NAND_ADQ14,EMAC2_RXD2,TRACE_D2		HPS_IOB_23			A47				
HPS		HPS_IOB_24	GPIO1_IO23,SPIM0_SS0_N,SPIS1_MISO,MDO10_MDC,I2C_EMAC0_SCL,NAND_ADQ15,EMAC2_RXD3,TRACE_D3		HPS_IOB_24			B47				
3A_T		95_IO	IOB			HPS_DDR	DIFF_IO_3A_T1n	No	W2	DQ32	DQ16	DQ8

altera				Pin Information for Aglex™ 5 A5ED052A Device							
				Version: 2025-06-06							
				B23A Status: Final							
Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B23A	DQS for X4	DQS for X8/X9	DQS for X16/X18
3A.T	94	IO	IOB		HPS_DDR	DIFF_IO_3A.T1p	No	W1	DQ32	DQ16	DQ8
3A.T	93	IO	IOB		HPS_DDR	DIFF_IO_3A.T2n	No	T2	DQ32	DQ16	DQ8
3A.T	92	IO	IOB		HPS_DDR	DIFF_IO_3A.T2p	No	N1	DQ32	DQ16	DQ8
3A.T	91	IO	IOB		HPS_DDR	DIFF_IO_3A.T3n	No	N2	DQSn32	DQ16	DQ8
3A.T	90	IO	IOB	AVST_READY	HPS_DDR	DIFF_IO_3A.T3p	No	M1	DQ32	DQ16	DQ8
3A.T	89	IO	IOB		HPS_DDR	DIFF_IO_3A.T4n	No	J1	DQSn33	DQSn16/CQn16	DQ8
3A.T	88	IO	IOB		HPS_DDR	DIFF_IO_3A.T4p	No	J2	DQ33	DQSn16/CQ16	DQ8
3A.T	87	IO	IOB		HPS_DDR	DIFF_IO_3A.T5n	Yes	C2	DQ33	DQ16	DQ8
3A.T	86	IO	IOB		HPS_DDR	DIFF_IO_3A.T5p	Yes	E2	DQ33	DQ16	DQ8
3A.T	85	IO	IOB		HPS_DDR	DIFF_IO_3A.T6n	No	G1	DQ33	DQ16	DQ8
3A.T	84	IO	IOB		HPS_DDR	DIFF_IO_3A.T6p	No	G2	DQ33	DQ16	DQ8
3A.T	83	IO	IOB	AVST_DATA10	HPS_DDR	DIFF_IO_3A.T7n	No	B4	DQ34	DQ17	DQ8
3A.T	82	IO	IOB	AVST_DATA9	HPS_DDR	DIFF_IO_3A.T7p	No	B8	DQ34	DQ17	DQ8
3A.T	81	IO	IOB	AVST_DATA8	HPS_DDR	DIFF_IO_3A.T8n	No	A16	DQ34	DQ17	DQ8
3A.T	80	IO	IOB	AVST_VALID	HPS_DDR	DIFF_IO_3A.T8p	No	B18	DQ34	DQ17	DQ8
3A.T	79	IO	IOB	AVST_DATA7	HPS_DDR	DIFF_IO_3A.T9n	No	B12	DQSn34	DQ17	DQ8
3A.T	78	IO	IOB	AVST_DATA6	HPS_DDR	DIFF_IO_3A.T9p	No	A12	DQ34	DQ17	DQ8
3A.T	77	IO	IOB	AVST_DATA5	HPS_DDR	DIFF_IO_3A.T10n	No	A18	DQSn35	DQSn17/CQn17	DQSn8/CQn8
3A.T	76	IO	IOB	AVST_DATA4	HPS_DDR	DIFF_IO_3A.T10p	No	B20	DQ35	DQSn17/CQ17	DQSn8/CQ8
3A.T	75	IO	IOB	AVST_DATA3	HPS_DDR	DIFF_IO_3A.T11n	Yes	A25	DQ35	DQ17	DQ8
3A.T	74	IO	IOB	AVST_DATA2	HPS_DDR	DIFF_IO_3A.T11p	Yes	B25	DQ35	DQ17	DQ8
3A.T	73	IO	IOB	AVST_DATA1	HPS_DDR	DIFF_IO_3A.T12n	No	A23	DQ35	DQ17	DQ8
3A.T	72	IO	IOB	AVST_DATA0	HPS_DDR	DIFF_IO_3A.T12p	No	B23	DQ35	DQ17	DQ8
3A.T	71	IO	IOB		HPS_DDR	DIFF_IO_3A.T13n	No	F11	DQ36	DQ18	DQ9
3A.T	70	IO	IOB		HPS_DDR	DIFF_IO_3A.T13p	No	D11	DQ36	DQ18	DQ9
3A.T	69	IO	IOB		HPS_DDR	DIFF_IO_3A.T14n	No	H15	DQ36	DQ18	DQ9
3A.T	68	IO	IOB		HPS_DDR	DIFF_IO_3A.T14p	No	F15	DQ36	DQ18	DQ9
3A.T	67	IO	IOB		HPS_DDR	DIFF_IO_3A.T15n	No	K15	DQSn36	DQ18	DQ9
3A.T	66	IO	IOB		HPS_DDR	DIFF_IO_3A.T15p	No	K11	DQ36	DQ18	DQ9
3A.T	65	IO	IOB,PLL_3A.T_CLKOUT1n		HPS_DDR	DIFF_IO_3A.T16n	No	D21	DQSn37	DQSn18/CQn18	DQ9
3A.T	64	IO	IOB,PLL_3A.T_CLKOUT1p,PLL_3A.T_CLKOUT1,PLL_3A.T_FB1		HPS_DDR	DIFF_IO_3A.T16p	No	F21	DQ37	DQSn18/CQ18	DQ9
3A.T	63	IO	IOB		HPS_DDR	DIFF_IO_3A.T17n	Yes	H24	DQ37	DQ18	DQ9
3A.T	62	IO	IOB,RZQ_T_3A		HPS_DDR	DIFF_IO_3A.T17p	Yes	F24	DQ37	DQ18	DQ9
3A.T	61	IO	IOB,CLK_T_3A_1n		HPS_DDR	DIFF_IO_3A.T18n	No	K24	DQ37	DQ18	DQ9
3A.T	60	IO	IOB,CLK_T_3A_1p		HPS_DDR	DIFF_IO_3A.T18p	No	K21	DQ37	DQ18	DQ9
3A.T	59	IO	IOB,CLK_T_3A_0n		HPS_DDR	DIFF_IO_3A.T19n	No	V22	DQ38	DQ19	DQ9
3A.T	58	IO	IOB,CLK_T_3A_0p		HPS_DDR	DIFF_IO_3A.T19p	No	P22	DQ38	DQ19	DQ9
3A.T	57	IO	IOB		HPS_DDR	DIFF_IO_3A.T20n	No	P25	DQ38	DQ19	DQ9
3A.T	56	IO	IOB		HPS_DDR	DIFF_IO_3A.T20p	No	P27	DQ38	DQ19	DQ9
3A.T	55	IO	IOB,PLL_3A.T_CLKOUT0n		HPS_DDR	DIFF_IO_3A.T21n	No	V27	DQSn38	DQ19	DQ9
3A.T	54	IO	IOB,PLL_3A.T_CLKOUT0p,PLL_3A.T_CLKOUT0,PLL_3A.T_FB0		HPS_DDR	DIFF_IO_3A.T21p	No	V30	DQ38	DQ19	DQ9
3A.T	53	IO	IOB	AVST_CLK	HPS_DDR	DIFF_IO_3A.T22n	No	V33	DQSn39	DQSn19/CQn19	DQSn9/CQn9
3A.T	52	IO	IOB	AVST_DATA15	HPS_DDR	DIFF_IO_3A.T22p	No	P33	DQ39	DQSn19/CQ19	DQSn9/CQ9
3A.T	51	IO	IOB	AVST_DATA14	HPS_DDR	DIFF_IO_3A.T23n	Yes	P35	DQ39	DQ19	DQ9
3A.T	50	IO	IOB	AVST_DATA13	HPS_DDR	DIFF_IO_3A.T23p	Yes	V35	DQ39	DQ19	DQ9
3A.T	49	IO	IOB	AVST_DATA12	HPS_DDR	DIFF_IO_3A.T24n	No	P39	DQ39	DQ19	DQ9
3A.T	48	IO	IOB	AVST_DATA11	HPS_DDR	DIFF_IO_3A.T24p	No	V41	DQ39	DQ19	DQ9
3A.B	47	IO	IOB		HPS_DDR	DIFF_IO_3A.B1n	No	D28	DQ40	DQ20	DQ10
3A.B	46	IO	IOB		HPS_DDR	DIFF_IO_3A.B1p	No	F28	DQ40	DQ20	DQ10
3A.B	45	IO	IOB		HPS_DDR	DIFF_IO_3A.B2n	No	F31	DQ40	DQ20	DQ10
3A.B	44	IO	IOB		HPS_DDR	DIFF_IO_3A.B2p	No	H31	DQ40	DQ20	DQ10
3A.B	43	IO	IOB		HPS_DDR	DIFF_IO_3A.B3n	No	K28	DQSn40	DQ20	DQ10
3A.B	42	IO	IOB		HPS_DDR	DIFF_IO_3A.B3p	No	K31	DQ40	DQ20	DQ10
3A.B	41	IO	IOB,PLL_3A.B_CLKOUT1n		HPS_DDR	DIFF_IO_3A.B4n	No	D36	DQSn41	DQSn20/CQn20	DQ10
3A.B	40	IO	IOB,PLL_3A.B_CLKOUT1p,PLL_3A.B_CLKOUT1,PLL_3A.B_FB1		HPS_DDR	DIFF_IO_3A.B4p	No	F36	DQ41	DQSn20/CQ20	DQ10
3A.B	39	IO	IOB		HPS_DDR	DIFF_IO_3A.B5n	Yes	K36	DQ41	DQ20	DQ10
3A.B	38	IO	IOB,RZQ_B_3A		HPS_DDR	DIFF_IO_3A.B5p	Yes	K38	DQ41	DQ20	DQ10
3A.B	37	IO	IOB,CLK_B_3A_1n		HPS_DDR	DIFF_IO_3A.B6n	Yes	H38	DQ41	DQ20	DQ10
3A.B	36	IO	IOB,CLK_B_3A_1p		HPS_DDR	DIFF_IO_3A.B6p	Yes	F38	DQ41	DQ20	DQ10
3A.B	35	IO	IOB,CLK_B_3A_0n		HPS_DDR	DIFF_IO_3A.B7n	No	B29	DQ42	DQ21	DQ10
3A.B	34	IO	IOB,CLK_B_3A_0p		HPS_DDR	DIFF_IO_3A.B7p	No	A26	DQ42	DQ21	DQ10
3A.B	33	IO	IOB		HPS_DDR	DIFF_IO_3A.B8n	No	A29	DQ42	DQ21	DQ10
3A.B	32	IO	IOB		HPS_DDR	DIFF_IO_3A.B8p	No	B32	DQ42	DQ21	DQ10
3A.B	31	IO	IOB,PLL_3A.B_CLKOUT0n		HPS_DDR	DIFF_IO_3A.B9n	No	A34	DQSn42	DQ21	DQ10
3A.B	30	IO	IOB,PLL_3A.B_CLKOUT0p,PLL_3A.B_CLKOUT0,PLL_3A.B_FB0		HPS_DDR	DIFF_IO_3A.B9p	No	B34	DQ42	DQ21	DQ10
3A.B	29	IO	IOB		HPS_DDR	DIFF_IO_3A.B10n	No	B37	DQSn43	DQSn21/CQn21	DQSn10/CQn10
3A.B	28	IO	IOB		HPS_DDR	DIFF_IO_3A.B10p	No	A37	DQ43	DQSn21/CQ21	DQSn10/CQ10
3A.B	27	IO	IOB		HPS_DDR	DIFF_IO_3A.B11n	Yes	A42	DQ43	DQ21	DQ10
3A.B	26	IO	IOB		HPS_DDR	DIFF_IO_3A.B11p	Yes	B44	DQ43	DQ21	DQ10
3A.B	25	IO	IOB		HPS_DDR	DIFF_IO_3A.B12n	Yes	A40	DQ43	DQ21	DQ10
3A.B	24	IO	IOB		HPS_DDR	DIFF_IO_3A.B12p	Yes	B42	DQ43	DQ21	DQ10
3A.B	23	IO	IOB		HPS_DDR	DIFF_IO_3A.B13n	No	D43	DQ44	DQ22	DQ11
3A.B	22	IO	IOB		HPS_DDR	DIFF_IO_3A.B13p	No	F43	DQ44	DQ22	DQ11
3A.B	21	IO	IOB		HPS_DDR	DIFF_IO_3A.B14n	No	F46	DQ44	DQ22	DQ11
3A.B	20	IO	IOB		HPS_DDR	DIFF_IO_3A.B14p	No	H46	DQ44	DQ22	DQ11
3A.B	19	IO	IOB		HPS_DDR	DIFF_IO_3A.B15n	No	K43	DQSn44	DQ22	DQ11
3A.B	18	IO	IOB		HPS_DDR	DIFF_IO_3A.B15p	No	K46	DQ44	DQ22	DQ11
3A.B	17	IO	IOB		HPS_DDR	DIFF_IO_3A.B16n	No	D52	DQSn45	DQSn22/CQn22	DQ11
3A.B	16	IO	IOB		HPS_DDR	DIFF_IO_3A.B16p	No	F52	DQ45	DQSn22/CQ22	DQ11
3A.B	15	IO	IOB		HPS_DDR	DIFF_IO_3A.B17n	Yes	K52	DQ45	DQ22	DQ11
3A.B	14	IO	IOB		HPS_DDR	DIFF_IO_3A.B17p	Yes	K55	DQ45	DQ22	DQ11
3A.B	13	IO	IOB		HPS_DDR	DIFF_IO_3A.B18n	Yes	F55	DQ45	DQ22	DQ11

altera

Pin Information for Agilix™ 5 A5ED052A Device

Version: 2025-06-06

B23A Status: Final

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B23A	DQS for X4	DQS for X8/X9	DQS for X16/X18
3A_B	12	IO	IOB		HPS_DDR	DIFF_IO_3A_B18p	Yes	H55	DQ45	DQ22	DQ11
3A_B	11	IO	IOB		HPS_DDR	DIFF_IO_3A_B19n	No	V45	DQ46	DQ23	DQ11
3A_B	10	IO	IOB		HPS_DDR	DIFF_IO_3A_B19p	No	P41	DQ46	DQ23	DQ11
3A_B	9	IO	IOB		HPS_DDR	DIFF_IO_3A_B20n	No	P48	DQ46	DQ23	DQ11
3A_B	8	IO	IOB		HPS_DDR	DIFF_IO_3A_B20p	No	V48	DQ46	DQ23	DQ11
3A_B	7	IO	IOB		HPS_DDR	DIFF_IO_3A_B21n	No	P50	DQSn46	DQ23	DQ11
3A_B	6	IO	IOB		HPS_DDR	DIFF_IO_3A_B21p	No	V50	DQSn46	DQ23	DQ11
3A_B	5	IO	IOB		HPS_DDR	DIFF_IO_3A_B22n	No	P53	DQSn47	DQSn23/CQn23	DQSn11/CQn11
3A_B	4	IO	IOB		HPS_DDR	DIFF_IO_3A_B22p	No	P56	DQSn47	DQSn23/CQ23	DQSn11/CQ11
3A_B	3	IO	IOB		HPS_DDR	DIFF_IO_3A_B23n	Yes	V56	DQ47	DQ23	DQ11
3A_B	2	IO	IOB		HPS_DDR	DIFF_IO_3A_B23p	Yes	V58	DQ47	DQ23	DQ11
3A_B	1	IO	IOB		HPS_DDR	DIFF_IO_3A_B24n	Yes	P61	DQ47	DQ23	DQ11
3A_B	0	IO	IOB		HPS_DDR	DIFF_IO_3A_B24p	Yes	V61	DQ47	DQ23	DQ11
5A		IO	HVIO_5A_1, TXCLK1, Data_Ctrl1					BG81			
5A		IO	HVIO_5A_2, TXCLK2, Data_Ctrl2					BD81			
5A		IO	HVIO_5A_3, TXCLK3, Data_Ctrl3					BG80			
5A		IO	HVIO_5A_4, TXCLK4, Data_Ctrl4					BF78			
5A		IO	HVIO_5A_5, TXCLK5, Data_Ctrl5					BK80			
5A		IO	HVIO_5A_6, PIN_PERST_N_CVP_L1B_0, TXCLK6, Data_Ctrl6					BK81			
5A		IO	HVIO_5A_7, PIN_PERST_N_CVP_L1C_0, TXCLK7, Data_Ctrl7					BH74			
5A		IO	HVIO_5A_8, TXCLK8, Data_Ctrl8					BF74			
5A		IO	HVIO_5A_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					BL78			
5A		IO	HVIO_5A_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					BM80			
5A		IO	HVIO_5A_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					BP81			
5A		IO	HVIO_5A_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					BP80			
5A		IO	HVIO_5A_13, TXCLK13, Data_Ctrl13					BR78			
5A		IO	HVIO_5A_14, TXCLK14, Data_Ctrl14					BN78			
5A		IO	HVIO_5A_15, TXCLK15, Data_Ctrl15					BU78			
5A		IO	HVIO_5A_16, TXCLK16, Data_Ctrl16					BU80			
5A		IO	HVIO_5A_17, TXCLK17, Data_Ctrl17					BR74			
5A		IO	HVIO_5A_18, TXCLK18, Data_Ctrl18					BN74			
5A		IO	HVIO_5A_19, TXCLK19, Data_Ctrl19					BU69			
5A		IO	HVIO_5A_20, TXCLK20, Data_Ctrl20					BE69			
5B		IO	HVIO_5B_1, SYSPLLREFCLK_L1A_0, TXCLK1, Data_Ctrl1					BU57			
5B		IO	HVIO_5B_2, SYSPLLREFCLK_L1A_1, TXCLK2, Data_Ctrl2					BN62			
5B		IO	HVIO_5B_3, SYSPLLREFCLK_L1B_0, TXCLK3, Data_Ctrl3					BN60			
5B		IO	HVIO_5B_4, SYSPLLREFCLK_L1B_1, TXCLK4, Data_Ctrl4					BR60			
5B		IO	HVIO_5B_5, TXCLK5, Data_Ctrl5					BV59			
5B		IO	HVIO_5B_6, PIN_PERST_N_CVP_L1B_1, TXCLK6, Data_Ctrl6					BU61			
5B		IO	HVIO_5B_7, PIN_PERST_N_CVP_L1C_1, TXCLK7, Data_Ctrl7					BU59			
5B		IO	HVIO_5B_8, TXCLK8, Data_Ctrl8					BV61			
5B		IO	HVIO_5B_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					BV63			
5B		IO	HVIO_5B_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					BU65			
5B		IO	HVIO_5B_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					BN67			
5B		IO	HVIO_5B_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					BR67			
5B		IO	HVIO_5B_13, TXCLK13, Data_Ctrl13					BU68			
5B		IO	HVIO_5B_14, TXCLK14, Data_Ctrl14					BV65			
5B		IO	HVIO_5B_15, TXCLK15, Data_Ctrl15					BN70			
5B		IO	HVIO_5B_16, TXCLK16, Data_Ctrl16					BU71			
5B		IO	HVIO_5B_17, TXCLK17, Data_Ctrl17					BV71			
5B		IO	HVIO_5B_18, TXCLK18, Data_Ctrl18					BV73			
5B		IO	HVIO_5B_19, SYSPLLREFCLK_L1C_0, TXCLK19, Data_Ctrl19					BV76			
5B		IO	HVIO_5B_20, TXCLK20, Data_Ctrl20					BU73			
6A		IO	HVIO_6A_1, SYSPLLREFCLK_R4A_0, TXCLK1, Data_Ctrl1					BN24			
6A		IO	HVIO_6A_2, SYSPLLREFCLK_R4A_1, TXCLK2, Data_Ctrl2					BN21			
6A		IO	HVIO_6A_3, SYSPLLREFCLK_R4B_0, TXCLK3, Data_Ctrl3					BU25			
6A		IO	HVIO_6A_4, SYSPLLREFCLK_R4B_1, TXCLK4, Data_Ctrl4					BV25			
6A		IO	HVIO_6A_5, TXCLK5, Data_Ctrl5					BU19			
6A		IO	HVIO_6A_6, TXCLK6, Data_Ctrl6					BN15			
6A		IO	HVIO_6A_7, PIN_PERST_N_R4C_0, TXCLK7, Data_Ctrl7					BR21			
6A		IO	HVIO_6A_8, TXCLK8, Data_Ctrl8					BL15			
6A		IO	HVIO_6A_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					BH15			
6A		IO	HVIO_6A_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					BF11			
6A		IO	HVIO_6A_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					BF15			
6A		IO	HVIO_6A_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					BH11			
6A		IO	HVIO_6A_13, TXCLK13, Data_Ctrl13					BN4			
6A		IO	HVIO_6A_14, TXCLK14, Data_Ctrl14					BN11			
6A		IO	HVIO_6A_15, TXCLK15, Data_Ctrl15					BR11			
6A		IO	HVIO_6A_16, TXCLK16, Data_Ctrl16					BL4			
6A		IO	HVIO_6A_17, TXCLK17, Data_Ctrl17					BH4			
6A		IO	HVIO_6A_18, TXCLK18, Data_Ctrl18					BF4			
6A		IO	HVIO_6A_19, SYSPLLREFCLK_R4C_0, TXCLK19, Data_Ctrl19					BC9			
6A		IO	HVIO_6A_20, TXCLK20, Data_Ctrl20					BC13			
6B		IO	HVIO_6B_1, TXCLK1, Data_Ctrl1					BK1			
6B		IO	HVIO_6B_2, TXCLK2, Data_Ctrl2					BG1			
6B		IO	HVIO_6B_3, TXCLK3, Data_Ctrl3					BK2			
6B		IO	HVIO_6B_4, TXCLK4, Data_Ctrl4					BM1			
6B		IO	HVIO_6B_5, TXCLK5, Data_Ctrl5					BP1			
6B		IO	HVIO_6B_6, TXCLK6, Data_Ctrl6					BM2			
6B		IO	HVIO_6B_7, PIN_PERST_N_R4C_1, TXCLK7, Data_Ctrl7					BT2			
6B		IO	HVIO_6B_8, TXCLK8, Data_Ctrl8					BP2			
6B		IO	HVIO_6B_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					BU4			

altera

Pin Information for Agilex™ 5 A5ED052A Device

Version: 2025-06-06

B23A Status: Final

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B23A	DQS for X4	DQS for X8/X9	DQS for X16/X18	
6B		IO	HVIO_6B_10,PLLREFCLK2,TXCLK10,RXCLK2_Data_Ctrl10					BU2				
		IO	HVIO_6B_11,SOURCE_SYNC_CLK1,TXCLK11,RXCLK3_Data_Ctrl11					BU8				
		IO	HVIO_6B_12,SOURCE_SYNC_CLK2,TXCLK12,RXCLK4_Data_Ctrl12					BU8				
		IO	HVIO_6B_13,TXCLK13_Data_Ctrl13					BU12				
		IO	HVIO_6B_14,TXCLK14_Data_Ctrl14					BU12				
		IO	HVIO_6B_15,TXCLK15_Data_Ctrl15					BU18				
		IO	HVIO_6B_16,TXCLK16_Data_Ctrl16					BU16				
		IO	HVIO_6B_17,TXCLK17_Data_Ctrl17					BU20				
		IO	HVIO_6B_18,TXCLK18_Data_Ctrl18					BU18				
		IO	HVIO_6B_19,TXCLK19_Data_Ctrl19					BU23				
		IO	HVIO_6B_20,TXCLK20_Data_Ctrl20					BU20				
	6C		IO	HVIO_6C_1,TXCLK1_Data_Ctrl1					P19			
			IO	HVIO_6C_2,TXCLK2_Data_Ctrl2					Y22			
			IO	HVIO_6C_3,TXCLK3_Data_Ctrl3					V19			
			IO	HVIO_6C_4,TXCLK4_Data_Ctrl4					AC22			
			IO	HVIO_6C_5,TXCLK5_Data_Ctrl5					AB14			
			IO	HVIO_6C_6,TXCLK6_Data_Ctrl6					AF22			
			IO	HVIO_6C_7,TXCLK7_Data_Ctrl7					AF19			
			IO	HVIO_6C_8,TXCLK8_Data_Ctrl8					AC19			
			IO	HVIO_6C_9,PLLREFCLK1,TXCLK9,RXCLK1_Data_Ctrl9					AE14			
		IO	HVIO_6C_10,PLLREFCLK2,TXCLK10,RXCLK2_Data_Ctrl10					AB7				
		IO	HVIO_6C_11,SOURCE_SYNC_CLK1,TXCLK11,RXCLK3_Data_Ctrl11					AE5				
		IO	HVIO_6C_12,SOURCE_SYNC_CLK2,TXCLK12,RXCLK4_Data_Ctrl12					AE7				
		IO	HVIO_6C_13,TXCLK13_Data_Ctrl13					AK14				
		IO	HVIO_6C_14,TXCLK14_Data_Ctrl14					AB10				
		IO	HVIO_6C_15,TXCLK15_Data_Ctrl15					AK7				
		IO	HVIO_6C_16,TXCLK16_Data_Ctrl16					AM22				
		IO	HVIO_6C_17,TXCLK17_Data_Ctrl17					AJ22				
		IO	HVIO_6C_18,TXCLK18_Data_Ctrl18					AK10				
		IO	HVIO_6C_19,TXCLK19_Data_Ctrl19					AP19				
		IO	HVIO_6C_20,TXCLK20_Data_Ctrl20					AM19				
6D		IO	HVIO_6D_1,TXCLK1_Data_Ctrl1					AN1				
		IO	HVIO_6D_2,TXCLK2_Data_Ctrl2					AN2				
		IO	HVIO_6D_3,TXCLK3_Data_Ctrl3					AG1				
		IO	HVIO_6D_4,TXCLK4_Data_Ctrl4					AG2				
		IO	HVIO_6D_5,TXCLK5_Data_Ctrl5					AL1				
		IO	HVIO_6D_6,TXCLK6_Data_Ctrl6					AL2				
		IO	HVIO_6D_7,TXCLK7_Data_Ctrl7					AD2				
		IO	HVIO_6D_8,TXCLK8_Data_Ctrl8					AH2				
		IO	HVIO_6D_9,PLLREFCLK1,TXCLK9,RXCLK1_Data_Ctrl9					AA1				
		IO	HVIO_6D_10,PLLREFCLK2,TXCLK10,RXCLK2_Data_Ctrl10					AD1				
		IO	HVIO_6D_11,SOURCE_SYNC_CLK1,TXCLK11,RXCLK3_Data_Ctrl11					AA2				
		IO	HVIO_6D_12,SOURCE_SYNC_CLK2,TXCLK12,RXCLK4_Data_Ctrl12					D4				
		IO	HVIO_6D_13,TXCLK13_Data_Ctrl13					U7				
		IO	HVIO_6D_14,TXCLK14_Data_Ctrl14					U5				
		IO	HVIO_6D_15,TXCLK15_Data_Ctrl15					R7				
		IO	HVIO_6D_16,TXCLK16_Data_Ctrl16					F4				
		IO	HVIO_6D_17,TXCLK17_Data_Ctrl17					R10				
		IO	HVIO_6D_18,TXCLK18_Data_Ctrl18					K4				
		IO	HVIO_6D_19,TXCLK19_Data_Ctrl19					R14				
		IO	HVIO_6D_20,TXCLK20_Data_Ctrl20					U14				
1B		GTSL1B_RX_CH0p						BA81				
		GTSL1B_RX_CH0n						BA79				
		GTSL1B_RX_CH1p						AW81				
		GTSL1B_RX_CH1n						AW79				
		GTSL1B_RX_CH2p						AU81				
		GTSL1B_RX_CH2n						AU79				
		GTSL1B_RX_CH3p						AP81				
		GTSL1B_RX_CH3n						AR79				
		GTSL1B_TX_CH0p						BB75				
		GTSL1B_TX_CH0n						BB72				
		GTSL1B_TX_CH1p						AY75				
		GTSL1B_TX_CH1n						AY72				
		GTSL1B_TX_CH2p						AV75				
		GTSL1B_TX_CH2n						AV72				
		GTSL1B_TX_CH3p						AT75				
		GTSL1B_TX_CH3n						AT72				
		REFCLK_GTSL1B_CH1p						AW69				
		REFCLK_GTSL1B_CH1n						AW66				
		REFCLK_GTSL1B_RX_P						BA66				
		REFCLK_GTSL1B_RX_N						BA69				
1C		GTSL1C_RX_CH0p						AM81				
		GTSL1C_RX_CH0n						AM79				
		GTSL1C_RX_CH1p						AF81				
		GTSL1C_RX_CH1n						AF79				
		GTSL1C_RX_CH2p						Y81				
		GTSL1C_RX_CH2n						Y79				
		GTSL1C_RX_CH3p						P81				
		GTSL1C_RX_CH3n						P79				
		GTSL1C_TX_CH0p						AP75				
		GTSL1C_TX_CH0n						AP72				
		GTSL1C_TX_CH1p						AJ75				

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B23A	DQS for X4	DQS for X8/X9	DQS for X16/X18
1C		GTS1C_TX_CH1n						AJ72			
1C		GTS1C_TX_CH2p						AC75			
1C		GTS1C_TX_CH2n						AC72			
1C		GTS1C_TX_CH3p						V75			
1C		GTS1C_TX_CH3n						V72			
1C		REFCLK_GTS1C_CH1p						AR69			
1C		REFCLK_GTS1C_CH1n						AR66			
1C		CDRCLKOUT_GTS1C_CH2p						AM66			
1C		CDRCLKOUT_GTS1C_CH2n						AM69			
1C		REFCLK_GTS1C_RX_P						AU69			
1C		REFCLK_GTS1C_RX_N						AU66			
4C		GTSR4C_RX_CH0p						BB1			
4C		GTSR4C_RX_CH0n						BB3			
4C		GTSR4C_RX_CH1p						AV1			
4C		GTSR4C_RX_CH1n						AV3			
4C		GTSR4C_RX_CH2p						AV1			
4C		GTSR4C_RX_CH2n						AV3			
4C		GTSR4C_RX_CH3p						AT1			
4C		GTSR4C_RX_CH3n						AT3			
4C		GTSR4C_TX_CH0p						BA9			
4C		GTSR4C_TX_CH0n						BA13			
4C		GTSR4C_TX_CH1p						AW9			
4C		GTSR4C_TX_CH1n						AW13			
4C		GTSR4C_TX_CH2p						AU9			
4C		GTSR4C_TX_CH2n						AU13			
4C		GTSR4C_TX_CH3p						AR9			
4C		GTSR4C_TX_CH3n						AR13			
4C		REFCLK_GTSR4C_CH1p						AV19			
4C		REFCLK_GTSR4C_CH1n						AV17			
4C		CDRCLKOUT_GTSR4C_CH2p						AT19			
4C		CDRCLKOUT_GTSR4C_CH2n						AT17			
4C		REFCLK_GTSR4C_RX_P						AV17			
4C		REFCLK_GTSR4C_RX_N						AV19			
		GND						BV54			
		GND						BV49			
		GND						BU49			
		GND						BC33			
		GND						BE33			
		GND						AJ39			
		GND						Y77			
		GND						Y75			
		GND						Y72			
		GND						Y69			
		GND						Y66			
		GND						Y64			
		GND						Y61			
		GND						Y58			
		GND						Y53			
		GND						Y45			
		GND						Y33			
		GND						Y27			
		GND						Y19			
		GND						V81			
		GND						V79			
		GND						V77			
		GND						V69			
		GND						V53			
		GND						V39			
		GND						V25			
		GND						U10			
		GND						T1			
		GND						R5			
		GND						P77			
		GND						P75			
		GND						P72			
		GND						P69			
		GND						P58			
		GND						P45			
		GND						P30			
		GND						M2			
		GND						L81			
		GND						L79			
		GND						L77			
		GND						H74			
		GND						H67			
		GND						H60			
		GND						H52			
		GND						H43			
		GND						H4			
		GND						H36			
		GND						H28			
		GND						H21			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B23A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						H11			
		GND						E1			
		GND						D78			
		GND						D70			
		GND						D62			
		GND						D55			
		GND						D46			
		GND						D38			
		GND						D31			
		GND						D24			
		GND						D15			
		GND						C81			
		GND						C1			
		GND						BV80			
		GND						BV78			
		GND						BV68			
		GND						BV57			
		GND						BV44			
		GND						BV4			
		GND						BV34			
		GND						BV23			
		GND						BV2			
		GND						BU81			
		GND						BU76			
		GND						BU63			
		GND						BU51			
		GND						BU40			
		GND						BU26			
		GND						BU16			
		GND						BU1			
		GND						BT81			
		GND						BT80			
		GND						BT1			
		GND						BR70			
		GND						BR62			
		GND						BR55			
		GND						BR46			
		GND						BR4			
		GND						BR38			
		GND						BR31			
		GND						BR24			
		GND						BR15			
		GND						BM81			
		GND						BL74			
		GND						BL11			
		GND						BJ66			
		GND						BJ50			
		GND						BJ35			
		GND						BJ25			
		GND						BH78			
		GND						BG2			
		GND						BE61			
		GND						BE33			
		GND						BE45			
		GND						BE35			
		GND						BD80			
		GND						BC77			
		GND						BC75			
		GND						BC72			
		GND						BC69			
		GND						BC64			
		GND						BC6			
		GND						BC48			
		GND						BC39			
		GND						BC3			
		GND						BC27			
		GND						BC25			
		GND						BC22			
		GND						BC19			
		GND						BC17			
		GND						BC1			
		GND						BB9			
		GND						BB81			
		GND						BB79			
		GND						BB77			
		GND						BB69			
		GND						BB66			
		GND						BB64			
		GND						BB61			
		GND						BB6			
		GND						BB58			
		GND						BB56			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B23A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						B050			
		GND						B041			
		GND						B033			
		GND						B027			
		GND						B017			
		GND						B013			
		GND						BA77			
		GND						BA75			
		GND						BA72			
		GND						BA64			
		GND						BA6			
		GND						BA56			
		GND						BA53			
		GND						BA45			
		GND						BA35			
		GND						BA3			
		GND						BA27			
		GND						BA22			
		GND						BA19			
		GND						BA17			
		GND						BA1			
		GND						B01			
		GND						B03			
		GND						B51			
		GND						B40			
		GND						B26			
		GND						B2			
		GND						B16			
		GND						AY9			
		GND						AY81			
		GND						AY79			
		GND						AY77			
		GND						AY69			
		GND						AY66			
		GND						AY64			
		GND						AY61			
		GND						AY6			
		GND						AY58			
		GND						AY56			
		GND						AY48			
		GND						AY39			
		GND						AY30			
		GND						AY27			
		GND						AY22			
		GND						AY13			
		GND						AW77			
		GND						AW75			
		GND						AW72			
		GND						AW64			
		GND						AW61			
		GND						AW6			
		GND						AV56			
		GND						AV50			
		GND						AW41			
		GND						AW33			
		GND						AW3			
		GND						AW25			
		GND						AW19			
		GND						AW17			
		GND						AW1			
		GND						AV9			
		GND						AV81			
		GND						AV79			
		GND						AV77			
		GND						AV69			
		GND						AV66			
		GND						AV6			
		GND						AV53			
		GND						AV45			
		GND						AV35			
		GND						AV25			
		GND						AV13			
		GND						AU77			
		GND						AU75			
		GND						AU72			
		GND						AU64			
		GND						AU6			
		GND						AU56			
		GND						AU46			
		GND						AU39			
		GND						AU30			
		GND						AU3			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B23A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						AJ19			
		GND						AJ17			
		GND						AJ1			
		GND						AT9			
		GND						AT81			
		GND						AT79			
		GND						AT77			
		GND						AT69			
		GND						AT66			
		GND						AT61			
		GND						AT6			
		GND						AT50			
		GND						AT33			
		GND						AT13			
		GND						AR77			
		GND						AR75			
		GND						AR72			
		GND						AR64			
		GND						AR6			
		GND						AR58			
		GND						AR53			
		GND						AR45			
		GND						AR35			
		GND						AR3			
		GND						AR27			
		GND						AR25			
		GND						AR22			
		GND						AR19			
		GND						AR17			
		GND						AR1			
		GND						AP9			
		GND						AP81			
		GND						AP79			
		GND						AP77			
		GND						AP69			
		GND						AP66			
		GND						AP6			
		GND						AP56			
		GND						AP48			
		GND						AP39			
		GND						AP30			
		GND						AP17			
		GND						AP13			
		GND						AM77			
		GND						AM75			
		GND						AM72			
		GND						AM53			
		GND						AM50			
		GND						AM41			
		GND						AM33			
		GND						AM25			
		GND						AM5			
		GND						AJ81			
		GND						AJ79			
		GND						AJ77			
		GND						AJ69			
		GND						AJ66			
		GND						AJ61			
		GND						AJ45			
		GND						AJ35			
		GND						AJ27			
		GND						AJ19			
		GND						AH1			
		GND						AF77			
		GND						AF75			
		GND						AF72			
		GND						AF69			
		GND						AF64			
		GND						AF56			
		GND						AF48			
		GND						AF39			
		GND						AF30			
		GND						AF25			
		GND						AE10			
		GND						AC81			
		GND						AC79			
		GND						AC77			
		GND						AC69			
		GND						AC56			
		GND						AC50			
		GND						AC41			
		GND						AC35			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B23A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						A85			
		GND						A80			
		GND						A8			
		GND						A78			
		GND						A68			
		GND						A57			
		GND						A44			
		GND						A4			
		GND						A32			
		GND						A20			
		GNDSENSE						AT41			
		VCC						AY53			
		VCC						AY50			
		VCC						AY45			
		VCC						AY41			
		VCC						AY35			
		VCC						AY33			
		VCC						AW53			
		VCC						AW48			
		VCC						AW45			
		VCC						AW39			
		VCC						AW35			
		VCC						AW30			
		VCC						AV50			
		VCC						AV48			
		VCC						AV41			
		VCC						AV39			
		VCC						AV33			
		VCC						AV30			
		VCC						AU53			
		VCC						AU50			
		VCC						AU45			
		VCC						AU41			
		VCC						AU35			
		VCC						AU33			
		VCC						AT53			
		VCC						AT48			
		VCC						AT39			
		VCC						AT35			
		VCC						AR50			
		VCC						AR48			
		VCC						AR41			
		VCC						AR39			
		VCC						AR33			
		VCC						AP53			
		VCC						AP50			
		VCC						AP45			
		VCC						AP41			
		VCC						AP35			
		VCC						AP33			
		VCC						AM48			
		VCC						AM45			
		VCCPT						BA50			
		VCCPT						BA48			
		VCCPT						BA41			
		VCCPT						AM35			
		VCCPT						AU33			
		VCCPT_HVIO						BC61			
		VCCPT_HVIO						BC58			
		VCCPT_HVIO						BC30			
		VCCPT_HVIO						BB30			
		VCCPT_HVIO						AJ25			
		VCCPT_HVIO						AF27			
		DNU						BR43			
		DNU						BU54			
		DNU						BU37			
		DNU						BV37			
		DNU						P66			
		DNU						V66			
		DNU						BB35			
		DNU						BA39			
		DNU						AC39			
		DNU						AM39			
		DNU						BE64			
		DNU						B/22			
		DNU						AM27			
		DNU						AC61			
		DNU						AU25			
		DNU						BE66			
		DNU						BE19			
		DNU						AP25			
		DNU						BC66			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B23A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		DNV						BE22			
		VCCBAT						AF27			
		VCCIO_HPS						BE46			
		VCCIO_HPS						AJ50			
		VCCIO_HVIO_5A						AF50			
		VCCIO_HVIO_5A						BJ64			
		VCCIO_HVIO_5B						BJ61			
		VCCIO_HVIO_5B						BE58			
		VCCIO_HVIO_5B						BE56			
		VCCIO_HVIO_6A						BJ27			
		VCCIO_HVIO_6A						BE30			
		VCCIO_HVIO_6B						BE27			
		VCCIO_HVIO_6B						BE25			
		VCCIO_HVIO_6C						Y30			
		VCCIO_HVIO_6C						AC27			
		VCCIO_HVIO_6D						Y25			
		VCCIO_HVIO_6D						AC25			
		VCCIO_PIO_3A_B						Y41			
		VCCIO_PIO_3A_B						Y39			
		VCCIO_PIO_3A_T						Y35			
		VCCIO_PIO_3A_T						AC33			
		VCCIO_PIO_SDM						BB39			
		VCCIO_SDM						BE39			
		NC						AP22			
		NC						AM30			
		NC						AT30			
		NC						AR30			
		VCCRCORE						AV56			
		VCCRCORE						AT56			
		VCCRCORE						AR56			
		VCCADC						BC53			
		VCCEHT_GTSL1B						AT64			
		VCCEHT_GTSL1B						AR61			
		VCCEHT_GTSL1C						AJ64			
		VCCEHT_GTSL1C						AF61			
		VCCEHT_GTSR4C						AT25			
		VCCEHT_GTSR4C						AT22			
		VCCERT_GTSL1B						AV64			
		VCCERT_GTSL1B						AV61			
		VCCERT_GTSL1B						AU61			
		VCCERT_GTSL1C						AP64			
		VCCERT_GTSL1C						AP61			
		VCCERT_GTSL1C						AM64			
		VCCERT_GTSR4C						AW22			
		VCCERT_GTSR4C						AV22			
		VCCERT_GTSR4C						AU22			
		VCCFUSEWR_SDM						BE50			
		VCCH_SDM						BC56			
		VCCLSENSE						AT45			
		VCCL_ADC_SDM						BC35			
		VCCL_HPS						AJ56			
		VCCL_HPS						AJ53			
		VCCL_HPS						AF53			
		VCCL_HPS						AC53			
		VCCL_HPS_CORE0_CORE1						AC48			
		VCCL_HPS_CORE0_CORE1						AC45			
		VCCL_HPS_CORE2						Y56			
		VCCL_HPS_CORE2						AC56			
		VCCL_HPS_CORE3						Y50			
		VCCL_HPS_CORE3						Y48			
		VCCL_SDM						BC45			
		VCCL_SDM						BC41			
		VCCL_SDM						BB48			
		VCCL_SDM						BB45			
		VCCP						BA33			
		VCCP						BA30			
		VCCP						AF41			
		VCCP						AF35			
		VCCP						AF33			
		VCCP						AC30			
		VCCPLL1_HPS						AJ41			
		VCCPLL2_HPS						AF45			
		VCCPLLDIG1_HPS						AM56			
		VCCPLLDIG2_HPS						AJ48			
		VCCPLLDIG_SDM						BC50			
		VCCPLL_SDM						BB53			
		VCC_HSSI_L1						AW58			
		VCC_HSSI_L1						AV58			
		VCC_HSSI_L1						AU58			
		VCC_HSSI_L1						AT58			
		VCC_HSSI_L1						AP58			

altera Pin Information for Agilex™ 5 A5ED052A Device Version: 2025-06-06 B23A Status: Final											
Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B23A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		VCC_HSSI_L1						AM58			
		VCC_HSSI_L1						AF58			
		VCC_HSSI_R4						AW27			
		VCC_HSSI_R4						AV27			
		VCC_HSSI_R4						AU27			
		VCC_HSSI_R4						AT27			
		VCC_IO_SDM						BE41			
		APROBE2_GTSL1C						AC66			
		APROBE2_GTSR4C						BB22			
		APROBE_GTSL1B_CH3						AM61			
		APROBE_GTSL1C_CH2						AF66			
		APROBE_GTSL1C_CH3						AC64			
		APROBE_GTSR4C_CH2						BB19			
		APROBE_GTSR4C_CH3						AY25			
		RCOMP_GTSL1C_N						BA61			
		RCOMP_GTSL1C_P						BA58			
		RCOMP_GTSR4C_N						BB25			
		RCOMP_GTSR4C_P						BA25			

Pin Information for Agilex™ 5 A5ED052A Device											
Version: 2025-06-06											
B32A Status: Final											
Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
SDM		TDO						BW109			
SDM		TMS						CA112			
SDM		TCK						CA109			
SDM		TDI						BW112			
SDM		OSC_CLK_1						BR102			
SDM		SDM_IO1	AVStx8_DATA2,AS_DATA1					BK102			
SDM		SDM_IO5	AS_nCS00,MSEL0					CF112			
SDM		SDM_IO3	AVStx8_DATA3,AS_DATA2					CH99			
SDM		nCONFIG						BU99			
SDM		SDM_IO4	AVStx8_DATA1,AS_DATA0					BH99			
SDM		SDM_IO2	AVStx8_DATA0,AS_CLK					BK99			
SDM		SDM_IO7	AS_nCS02,MSEL1					BM99			
SDM		SDM_IO11	AVStx8_VALID,PWRMGMT_SDA					CF99			
SDM		nSTATUS						BW99			
SDM		SDM_IO16	PWRMGMT_SDA					BP102			
SDM		SDM_IO0	PWRMGMT_SCL,PWRMGMT_ALERT					CA99			
SDM		SDM_IO13	AVStx8_DATA5					BW102			
SDM		SDM_IO9	AS_nCS01,MSEL2,PWRMGMT_ALERT					BM102			
SDM		SDM_IO6	AVStx8_DATA4,AS_DATA3					CF102			
SDM		SDM_IO10	AVStx8_DATA7					CH109			
SDM		SDM_IO8	AVStx8_READY,AS_nCS03					CC102			
SDM		SDM_IO12	PWRMGMT_SDA,PWRMGMT_ALERT					BR99			
SDM		SDM_IO15	AVStx8_DATA6,AS_nRST					CA102			
SDM		SDM_IO14	AVStx8_CLK,PWRMGMT_SCL					CF109			
SDM		RREF_SDM						CL103			
SDM			TEMPODIODE0An					BF100			
SDM			TEMPODIODE0Ap					BE100			
SDM			VREFP_ADC					CL116			
SDM			VREFN_ADC					CK116			
SDM		VSIGP_0						CK119			
SDM		VSIGN_0						CL119			
SDM		VSIGP_1						CK113			
SDM		VSIGN_1						CL113			
HPS		HPS_IOA_1	GPIO0_IO0,SPIM0_SS1,N,SPIS0_CLK,UART0_CTS,N,NAND_ADQ0,SDMMC_DATA0,USB0_CLK,EMAC0_PP50,TRACE_D10		HPS_IOA_1			W135			
HPS		HPS_IOA_2	GPIO0_IO1,SPIM1_SS1,N,SPIS0_MOSI,UART0_RTS,N,NAND_ADQ1,SDMMC_DATA1,USB0_STP,EMAC0_PPSTRIG0,TRACE_D9		HPS_IOA_2			U135			
HPS		HPS_IOA_3	GPIO0_IO2,SPIS0_SS0,N,UART0_TX,I2C1_SDA,NAND_WE,N,SDMMC_CLK,USB0_DIR,EMAC1_PPS1,TRACE_D8		HPS_IOA_3			W134			
HPS		HPS_IOA_4	GPIO0_IO3,SPIS0_MISO,UART0_RX,I2C1_SCL,NAND_RE,N,USB0_DATA0,EMAC1_PPSTRIG1,TRACE_D7		HPS_IOA_4			AK115			
HPS		HPS_IOA_5	GPIO0_IO4,SPIM0_CLK,UART1_CTS,N,I2C0_SDA,NAND_WP,N,SDMMC_WRITE,PROTECT,USB0_DATA1,EMAC2_PPS2,TRACE_D6		HPS_IOA_5			U134			
HPS		HPS_IOA_6	GPIO0_IO5,SPIM0_MOSI,UART1_RTS,N,I2C0_SCL,NAND_ADQ2,SDMMC_DATA2,USB0_NXT,EMAC2_PPSTRIG2,TRACE_D5		HPS_IOA_6			AI120			
HPS		HPS_IOA_7	GPIO0_IO6,SPIM0_MISO,MADIO2_MDC,UART1_TX,I2C_EMAC2_SDA,NAND_ADQ3,SDMMC_DATA3,USB0_DATA2,TRACE_D4		HPS_IOA_7			R134			
HPS		HPS_IOA_8	GPIO0_IO7,SPIM0_SS0,N,MADIO2_MDC,UART1_RX,I2C_EMAC2_SCL,NAND_CLE,SDMMC_CMD,USB0_DATA3,TRACE_D15		HPS_IOA_8			AG115			
HPS		HPS_IOA_9	GPIO0_IO8,SPIM1_CLK,SPIS1_CLK,MADIO1_MDC,I2C_EMAC1_SCL,NAND_ADQ4,SDMMC_DATA4,USB0_DATA4,I3C1_SDA,TRACE_D14		HPS_IOA_9			N135			
HPS		HPS_IOA_10	GPIO0_IO9,SPIM1_MOSI,SPIS1_MOSI,MADIO1_MDC,I2C_EMAC1_SCL,NAND_ADQ5,SDMMC_DATA5,USB0_DATA5,I3C1_SCL,TRACE_D13		HPS_IOA_10			AK120			
HPS		HPS_IOA_11	GPIO0_IO10,SPIM1_MISO,SPIS1_SS0,N,MADIO0_MDC,I2C_EMAC0_SDA,NAND_ADQ6,SDMMC_DATA6,USB0_DATA6,I3C0_SDA,TRACE_D12		HPS_IOA_11			N134			
HPS		HPS_IOA_12	GPIO0_IO11,SPIM1_SS0,N,SPIS1_MISO,MADIO0_MDC,I2C_EMAC0_SCL,NAND_ADQ7,SDMMC_DATA7,USB0_DATA7,I3C0_SCL,TRACE_D11		HPS_IOA_12			T132			
HPS		HPS_IOA_13	GPIO0_IO12,NAND_ALE,SDMMC_PU_PD_DATA2,USB1_CLK,EMAC0_TX_CLK,TRACE_D10		HPS_IOA_13			P132			
HPS		HPS_IOA_14	GPIO0_IO13,NAND_RB,N,SDMMC_PWR_ENA,USB1_STP,EMAC0_TX_CTL,TRACE_D9		HPS_IOA_14			L135			
HPS		HPS_IOA_15	GPIO0_IO14,NAND_CE,N,USB1_DIR,EMAC0_RX_CLK,TRACE_D8		HPS_IOA_15			J135			
HPS		HPS_IOA_16	GPIO0_IO15,NAND_DQS,SDMMC_DATA_STROBE,USB1_DATA0,EMAC0_RX_CTL,TRACE_D7		HPS_IOA_16			AD135			
HPS		HPS_IOA_17	GPIO0_IO16,I3C1_SDA,NAND_ADQ8,USB1_DATA1,EMAC0_TXD0,TRACE_D6		HPS_IOA_17			M132			
HPS		HPS_IOA_18	GPIO0_IO17,I3C1_SCL,NAND_ADQ9,USB1_NXT,EMAC0_TXD1,TRACE_D5		HPS_IOA_18			AD134			
HPS		HPS_IOA_19	GPIO0_IO18,I3C0_SDA,NAND_ADQ10,USB1_DATA2,EMAC0_RXD0,TRACE_D4		HPS_IOA_19			K132			
HPS		HPS_IOA_20	GPIO0_IO19,SPIM1_SS1,N,I3C0_SCL,NAND_ADQ11,USB1_DATA3,EMAC0_RXD1,TRACE_CLK		HPS_IOA_20			AG129			
HPS		HPS_IOA_21	GPIO0_IO20,SPIM1_CLK,SPIS0_CLK,UART0_CTS,N,I2C1_SDA,NAND_ADQ12,USB1_DATA4,EMAC0_TXD2,TRACE_D0		HPS_IOA_21			J134			
HPS		HPS_IOA_22	GPIO0_IO21,SPIM1_MOSI,SPIS0_MOSI,UART0_RTS,N,I2C1_SCL,NAND_ADQ13,USB1_DATA5,EMAC0_TXD3,TRACE_D1		HPS_IOA_22			AG120			
HPS		HPS_IOA_23	GPIO0_IO22,SPIM1_MISO,SPIS0_SS0,N,UART0_TX,I2C0_SDA,NAND_ADQ14,USB1_DATA6,EMAC0_RXD2,TRACE_D2		HPS_IOA_23			G134			
HPS		HPS_IOA_24	GPIO0_IO23,SPIM1_SS0,N,SPIS0_MISO,UART0_RX,I2C0_SCL,NAND_ADQ15,USB1_DATA7,EMAC0_RXD3,TRACE_D3		HPS_IOA_24			G135			
HPS		HPS_IOB_1	GPIO1_IO0,SPIM1_CLK,UART0_CTS,N,EMAC0_PP50,NAND_ADQ0,SDMMC_DATA0,I3C0_SDA,PULLUP_EN,EMAC1_TX_CLK,TRACE_D10		HPS_IOB_1			E135			
HPS		HPS_IOB_2	GPIO1_IO1,SPIM1_MOSI,UART0_RTS,N,EMAC0_PPSTRIG0,NAND_ADQ1,SDMMC_DATA1,I3C1_SDA,PULLUP_EN,EMAC1_TX_CTL,TRACE_D9		HPS_IOB_2			F132			
HPS		HPS_IOB_3	GPIO1_IO2,SPIM1_MISO,UART0_TX,I2C0_SDA,NAND_WE,N,SDMMC_CLK,EMAC1_RX_CLK,TRACE_D8		HPS_IOB_3			D132			
HPS		HPS_IOB_4	GPIO1_IO3,SPIM1_SS0,N,UART0_RX,I2C0_SCL,NAND_RE,N,EMAC1_RX_CTL,TRACE_D7		HPS_IOB_4			AG123			
HPS		HPS_IOB_5	GPIO1_IO4,SPIM1_SS1,N,SPIS1_CLK,UART1_CTS,N,EMAC2_PPS2,NAND_WP,N,SDMMC_WRITE,PROTECT,I3C1_SDA,EMAC1_TXD0,TRACE_D6		HPS_IOB_5			B134			
HPS		HPS_IOB_6	GPIO1_IO5,SPIS1_MOSI,UART1_RTS,N,EMAC2_PPSTRIG2,NAND_ADQ2,SDMMC_DATA2,I3C1_SCL,EMAC1_TXD1,TRACE_D5		HPS_IOB_6			AA135			
HPS		HPS_IOB_7	GPIO1_IO6,SPIS1_SS0,N,UART1_TX,I2C1_SDA,NAND_ADQ3,SDMMC_DATA3,I3C0_SDA,EMAC1_RXD0,TRACE_D4		HPS_IOB_7			V127			
HPS		HPS_IOB_8	GPIO1_IO7,SPIS1_MISO,UART1_RX,I2C1_SCL,NAND_CLE,SDMMC_CMD,I3C0_SCL,EMAC1_RXD1,TRACE_D15		HPS_IOB_8			AB132			
HPS		HPS_IOB_9	GPIO1_IO8,ITAG_TCK,SPIS0_CLK,MADIO2_MDC,I2C_EMAC2_SDA,NAND_ADQ4,SDMMC_DATA4,EMAC1_TXD2,TRACE_D14		HPS_IOB_9			T127			
HPS		HPS_IOB_10	GPIO1_IO9,ITAG_TMS,SPIS0_MOSI,MADIO2_MDC,I2C_EMAC2_SCL,NAND_ADQ5,SDMMC_DATA5,EMAC1_TXD3,TRACE_D13		HPS_IOB_10			Y132			
HPS		HPS_IOB_11	GPIO1_IO10,ITAG_TDO,SPIS0_SS0,N,MADIO0_MDC,I2C_EMAC0_SDA,NAND_ADQ6,SDMMC_DATA6,EMAC1_RXD2,TRACE_D12		HPS_IOB_11			T124			
HPS		HPS_IOB_12	GPIO1_IO11,ITAG_TDI,SPIS0_MISO,MADIO0_MDC,I2C_EMAC0_SCL,NAND_ADQ7,SDMMC_DATA7,EMAC1_RXD3,TRACE_D11		HPS_IOB_12			P124			
HPS		HPS_IOB_13	GPIO1_IO12,I2C1_SDA,NAND_ALE,SDMMC_PU_PD_DATA2,EMAC2_TX_CLK,TRACE_D10		HPS_IOB_13			M127			
HPS		HPS_IOB_14	GPIO1_IO13,I2C1_SCL,NAND_RB,N,SDMMC_PWR_ENA,EMAC2_TX_CTL,TRACE_D9		HPS_IOB_14			K127			
HPS		HPS_IOB_15	GPIO1_IO14,UART1_TX,NAND_CE,N,I3C1_SDA,EMAC2_RX_CLK,TRACE_D8		HPS_IOB_15			M124			
HPS		HPS_IOB_16	GPIO1_IO15,UART1_RX,NAND_DQS,SDMMC_DATA_STROBE,I3C1_SCL,EMAC2_RX_CTL,TRACE_D7		HPS_IOB_16			AB127			
HPS		HPS_IOB_17	GPIO1_IO16,UART1_CTS,N,NAND_ADQ8,I3C0_SDA,EMAC2_TXD0,TRACE_D6		HPS_IOB_17			K124			
HPS		HPS_IOB_18	GPIO1_IO17,SPIM0_SS1,N,UART1_RTS,N,NAND_ADQ9,I3C0_SCL,EMAC2_TXD1,TRACE_D5		HPS_IOB_18			Y127			
HPS		HPS_IOB_19	GPIO1_IO18,SPIM0_MISO,MADIO1_MDC,I2C_EMAC1_SDA,NAND_ADQ10,EMAC2_RXD0,TRACE_D4		HPS_IOB_19			H127			
HPS		HPS_IOB_20	GPIO1_IO19,SPIM0_SS0,N,MADIO1_MDC,I2C_EMAC1_SCL,NAND_ADQ11,EMAC2_RXD1,TRACE_CLK		HPS_IOB_20			AB124			
HPS		HPS_IOB_21	GPIO1_IO20,SPIM0_CLK,SPIS1_CLK,I2C_EMAC2_SDA,NAND_ADQ12,I3C0_SDA,PULLUP_EN,EMAC2_TXD2,TRACE_D0		HPS_IOB_21			F127			
HPS		HPS_IOB_22	GPIO1_IO21,SPIM0_MOSI,SPIS1_MOSI,I2C_EMAC2_SCL,NAND_ADQ13,I3C1_SDA,PULLUP_EN,EMAC2_TXD3,TRACE_D1		HPS_IOB_22			Y124			
HPS		HPS_IOB_23	GPIO1_IO22,SPIM0_MISO,SPIS1_SS0,N,MADIO0_MDC,I2C_EMAC0_SDA,NAND_ADQ14,EMAC2_RXD2,TRACE_D2		HPS_IOB_23			F124			
HPS		HPS_IOB_24	GPIO1_IO23,SPIM0_SS0,N,SPIS1_MISO,MADIO0_MDC,I2C_EMAC0_SCL,NAND_ADQ15,EMAC2_RXD3,TRACE_D3		HPS_IOB_24			D124			
2A_T		95_IO	IOB			DIFF_IO_2A_T1n	No	CH59	DQ0	DQ0	DQ0

altera

Pin Information for Aglex™ 5 A5ED052A Device

Version: 2025-06-06

B32A Status: Final

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
2A, T	94 IO		IOB			DIFF_IO_2A_T1p	No	CF59	DQ0	DQ0	DQ0
2A, T	93 IO		IOB			DIFF_IO_2A_T2n	No	CH62	DQ0	DQ0	DQ0
2A, T	92 IO		IOB			DIFF_IO_2A_T2p	No	CF62	DQ0	DQ0	DQ0
2A, T	91 IO		IOB			DIFF_IO_2A_T3n	No	CC62	DQSn0	DQ0	DQ0
2A, T	90 IO		IOB			DIFF_IO_2A_T3p	No	CA62	DQSn0	DQ0	DQ0
2A, T	89 IO		IOB			DIFF_IO_2A_T4n	No	CF69	DQSn1	DQSn0/CQn0	DQ0
2A, T	88 IO		IOB			DIFF_IO_2A_T4p	No	CH69	DQS1	DQSn0/CQ0	DQ0
2A, T	87 IO		IOB			DIFF_IO_2A_T5n	Yes	CA71	DQ1	DQ0	DQ0
2A, T	86 IO		IOB			DIFF_IO_2A_T5p	Yes	CC71	DQ1	DQ0	DQ0
2A, T	85 IO		IOB			DIFF_IO_2A_T6n	No	CH71	DQ1	DQ0	DQ0
2A, T	84 IO		IOB			DIFF_IO_2A_T6p	No	CF71	DQ1	DQ0	DQ0
2A, T	83 IO		IOB			DIFF_IO_2A_T7n	No	CA59	DQ2	DQ1	DQ0
2A, T	82 IO		IOB			DIFF_IO_2A_T7p	No	BW59	DQ2	DQ1	DQ0
2A, T	81 IO		IOB			DIFF_IO_2A_T8n	No	BR59	DQ2	DQ1	DQ0
2A, T	80 IO		IOB			DIFF_IO_2A_T8p	No	BU59	DQ2	DQ1	DQ0
2A, T	79 IO		IOB			DIFF_IO_2A_T9n	No	BR62	DQSn2	DQ1	DQ0
2A, T	78 IO		IOB			DIFF_IO_2A_T9p	No	BU62	DQS2	DQ1	DQ0
2A, T	77 IO		IOB			DIFF_IO_2A_T10n	No	CA69	DQSn3	DQSn1/CQn1	DQSn0/CQn0
2A, T	76 IO		IOB			DIFF_IO_2A_T10p	No	BW69	DQS3	DQS1/CQ1	DQSn0/CQ0
2A, T	75 IO		IOB			DIFF_IO_2A_T11n	Yes	BU71	DQ3	DQ1	DQ0
2A, T	74 IO		IOB			DIFF_IO_2A_T11p	Yes	BR71	DQ3	DQ1	DQ0
2A, T	73 IO		IOB			DIFF_IO_2A_T12n	No	BU69	DQ3	DQ1	DQ0
2A, T	72 IO		IOB			DIFF_IO_2A_T12p	No	BR69	DQ3	DQ1	DQ0
2A, T	71 IO		IOB			DIFF_IO_2A_T13n	No	BK59	DQ4	DQ2	DQ1
2A, T	70 IO		IOB			DIFF_IO_2A_T13p	No	BM59	DQ4	DQ2	DQ1
2A, T	69 IO		IOB			DIFF_IO_2A_T14n	No	BH59	DQ4	DQ2	DQ1
2A, T	68 IO		IOB			DIFF_IO_2A_T14p	No	BH62	DQ4	DQ2	DQ1
2A, T	67 IO		IOB			DIFF_IO_2A_T15n	No	BP62	DQSn4	DQ2	DQ1
2A, T	66 IO		IOB			DIFF_IO_2A_T15p	No	BM62	DQS4	DQ2	DQ1
2A, T	65 IO		IOB,PLL_2A_T_CLKOUT1n			DIFF_IO_2A_T16n	No	BK69	DQSn5	DQSn2/CQn2	DQ1
2A, T	64 IO		IOB,PLL_2A_T_CLKOUT1p,PLL_2A_T_CLKOUT1,PLL_2A_T_FB1			DIFF_IO_2A_T16p	No	BM69	DQSn5	DQSn2/CQ2	DQ1
2A, T	63 IO		IOB			DIFF_IO_2A_T17n	Yes	BH71	DQ5	DQ2	DQ1
2A, T	62 IO		IOB,RZQ_T_2A			DIFF_IO_2A_T17p	Yes	BH69	DQ5	DQ2	DQ1
2A, T	61 IO		IOB,CLK_T_2A_1n			DIFF_IO_2A_T18n	No	BP71	DQ5	DQ2	DQ1
2A, T	60 IO		IOB,CLK_T_2A_1p			DIFF_IO_2A_T18p	No	BM71	DQ5	DQ2	DQ1
2A, T	59 IO		IOB,CLK_T_2A_0n			DIFF_IO_2A_T19n	No	BF72	DQ6	DQ3	DQ1
2A, T	58 IO		IOB,CLK_T_2A_0p			DIFF_IO_2A_T19p	No	BF75	DQ6	DQ3	DQ1
2A, T	57 IO		IOB			DIFF_IO_2A_T20n	No	BE75	DQ6	DQ3	DQ1
2A, T	56 IO		IOB			DIFF_IO_2A_T20p	No	BE79	DQ6	DQ3	DQ1
2A, T	55 IO		IOB,PLL_2A_T_CLKOUT0n			DIFF_IO_2A_T21n	No	BF83	DQSn6	DQ3	DQ1
2A, T	54 IO		IOB,PLL_2A_T_CLKOUT0p,PLL_2A_T_CLKOUT0,PLL_2A_T_FB0			DIFF_IO_2A_T21p	No	BE83	DQSn6	DQ3	DQ1
2A, T	53 IO		IOB			DIFF_IO_2A_T22n	No	BE86	DQSn7	DQSn3/CQn3	DQSn1/CQn1
2A, T	52 IO		IOB			DIFF_IO_2A_T22p	No	BF86	DQS7	DQSn3/CQ3	DQS1/CQ1
2A, T	51 IO		IOB			DIFF_IO_2A_T23n	Yes	BF90	DQ7	DQ3	DQ1
2A, T	50 IO		IOB			DIFF_IO_2A_T23p	Yes	BF93	DQ7	DQ3	DQ1
2A, T	49 IO		IOB			DIFF_IO_2A_T24n	No	BE93	DQ7	DQ3	DQ1
2A, T	48 IO		IOB			DIFF_IO_2A_T24p	No	BE96	DQ7	DQ3	DQ1
2A, B	47 IO		IOB			DIFF_IO_2A_B1n	No	BK78	DQ8	DQ4	DQ2
2A, B	46 IO		IOB			DIFF_IO_2A_B1p	No	BM78	DQ8	DQ4	DQ2
2A, B	45 IO		IOB			DIFF_IO_2A_B2n	No	BH78	DQ8	DQ4	DQ2
2A, B	44 IO		IOB			DIFF_IO_2A_B2p	No	BH81	DQ8	DQ4	DQ2
2A, B	43 IO		IOB			DIFF_IO_2A_B3n	No	BP81	DQSn8	DQ4	DQ2
2A, B	42 IO		IOB			DIFF_IO_2A_B3p	No	BM81	DQSn8	DQ4	DQ2
2A, B	41 IO		IOB,PLL_2A_B_CLKOUT1n			DIFF_IO_2A_B4n	No	BM89	DQSn9	DQSn4/CQn4	DQ2
2A, B	40 IO		IOB,PLL_2A_B_CLKOUT1p,PLL_2A_B_CLKOUT1,PLL_2A_B_FB1			DIFF_IO_2A_B4p	No	BK89	DQSn9	DQSn4/CQ4	DQ2
2A, B	39 IO		IOB			DIFF_IO_2A_B5n	Yes	BH92	DQ9	DQ4	DQ2
2A, B	38 IO		IOB,RZQ_B_2A			DIFF_IO_2A_B5p	Yes	BH89	DQ9	DQ4	DQ2
2A, B	37 IO		IOB,CLK_B_2A_1n			DIFF_IO_2A_B6n	Yes	BM92	DQ9	DQ4	DQ2
2A, B	36 IO		IOB,CLK_B_2A_1p			DIFF_IO_2A_B6p	Yes	BP92	DQ9	DQ4	DQ2
2A, B	35 IO		IOB,CLK_B_2A_0n			DIFF_IO_2A_B7n	No	CA78	DQ10	DQ5	DQ2
2A, B	34 IO		IOB,CLK_B_2A_0p			DIFF_IO_2A_B7p	No	BW78	DQ10	DQ5	DQ2
2A, B	33 IO		IOB			DIFF_IO_2A_B8n	No	BU78	DQ10	DQ5	DQ2
2A, B	32 IO		IOB			DIFF_IO_2A_B8p	No	BR78	DQ10	DQ5	DQ2
2A, B	31 IO		IOB,PLL_2A_B_CLKOUT0n			DIFF_IO_2A_B9n	No	BU81	DQSn10	DQ5	DQ2
2A, B	30 IO		IOB,PLL_2A_B_CLKOUT0p,PLL_2A_B_CLKOUT0,PLL_2A_B_FB0			DIFF_IO_2A_B9p	No	BR81	DQS10	DQ5	DQ2
2A, B	29 IO		IOB			DIFF_IO_2A_B10n	No	CA89	DQSn11	DQSn5/CQn5	DQSn2/CQn2
2A, B	28 IO		IOB			DIFF_IO_2A_B10p	No	BW89	DQS11	DQSn5/CQ5	DQS2/CQ2
2A, B	27 IO		IOB			DIFF_IO_2A_B11n	Yes	BU92	DQ11	DQ5	DQ2
2A, B	26 IO		IOB			DIFF_IO_2A_B11p	Yes	BR92	DQ11	DQ5	DQ2
2A, B	25 IO		IOB			DIFF_IO_2A_B12n	Yes	BU89	DQ11	DQ5	DQ2
2A, B	24 IO		IOB			DIFF_IO_2A_B12p	Yes	BR89	DQ11	DQ5	DQ2
2A, B	23 IO		IOB			DIFF_IO_2A_B13n	No	CF78	DQ12	DQ6	DQ3
2A, B	22 IO		IOB			DIFF_IO_2A_B13p	No	CH78	DQ12	DQ6	DQ3
2A, B	21 IO		IOB			DIFF_IO_2A_B14n	No	CC81	DQ12	DQ6	DQ3
2A, B	20 IO		IOB			DIFF_IO_2A_B14p	No	CA81	DQ12	DQ6	DQ3
2A, B	19 IO		IOB			DIFF_IO_2A_B15n	No	CH81	DQSn12	DQ6	DQ3
2A, B	18 IO		IOB			DIFF_IO_2A_B15p	No	CF81	DQS12	DQ6	DQ3
2A, B	17 IO		IOB			DIFF_IO_2A_B16n	No	CF89	DQSn13	DQSn6/CQn6	DQ3
2A, B	16 IO		IOB			DIFF_IO_2A_B16p	No	CH89	DQS13	DQSn6/CQ6	DQ3
2A, B	15 IO		IOB			DIFF_IO_2A_B17n	Yes	CH92	DQ13	DQ6	DQ3
2A, B	14 IO		IOB			DIFF_IO_2A_B17p	Yes	CF92	DQ13	DQ6	DQ3
2A, B	13 IO		IOB			DIFF_IO_2A_B18n	Yes	CA92	DQ13	DQ6	DQ3

altera

Pin Information for Aglex™ 5 A5ED052A Device

Version: 2025-06-06

B32A Status: Final

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
2A_B	12	IO	IOB			DIFF_IO_2A_B19p	Yes	CC92	DQ13	DQ6	DQ3
2A_B	11	IO	IOB			DIFF_IO_2A_B19n	No	CL76	DQ14	DQ7	DQ3
2A_B	10	IO	IOB			DIFF_IO_2A_B19p	No	CK76	DQ14	DQ7	DQ3
2A_B	9	IO	IOB			DIFF_IO_2A_B20n	No	CL82	DQ14	DQ7	DQ3
2A_B	8	IO	IOB			DIFF_IO_2A_B20p	No	CK80	DQ14	DQ7	DQ3
2A_B	7	IO	IOB			DIFF_IO_2A_B21n	No	CL85	DQSn14	DQ7	DQ3
2A_B	6	IO	IOB			DIFF_IO_2A_B21p	No	CK85	DQS14	DQ7	DQ3
2A_B	5	IO	IOB			DIFF_IO_2A_B22n	No	CK88	DQSn15	DQSn7/CQn7	DQSn3/CQn3
2A_B	4	IO	IOB			DIFF_IO_2A_B22p	No	CL88	DQS15	DQSn7/CQ7	DQSn3/CQ3
2A_B	3	IO	IOB			DIFF_IO_2A_B23n	Yes	CL97	DQ15	DQ7	DQ3
2A_B	2	IO	IOB			DIFF_IO_2A_B23p	Yes	CK97	DQ15	DQ7	DQ3
2A_B	1	IO	IOB			DIFF_IO_2A_B24n	Yes	CK94	DQ15	DQ7	DQ3
2A_B	0	IO	IOB			DIFF_IO_2A_B24p	Yes	CL91	DQ15	DQ7	DQ3
2B_T	95	IO	IOB			DIFF_IO_2B_T1n	No	CC19	DQ16	DQ8	DQ4
2B_T	94	IO	IOB			DIFF_IO_2B_T1p	No	CF19	DQ16	DQ8	DQ4
2B_T	93	IO	IOB			DIFF_IO_2B_T2n	No	CH22	DQ16	DQ8	DQ4
2B_T	92	IO	IOB			DIFF_IO_2B_T2p	No	CF22	DQ16	DQ8	DQ4
2B_T	91	IO	IOB			DIFF_IO_2B_T3n	No	CA22	DQSn16	DQ8	DQ4
2B_T	90	IO	IOB			DIFF_IO_2B_T3p	No	CC22	DQS16	DQ8	DQ4
2B_T	89	IO	IOB			DIFF_IO_2B_T4n	No	CC28	DQSn17	DQSn8/CQn8	DQ4
2B_T	88	IO	IOB			DIFF_IO_2B_T4p	No	CF28	DQS17	DQSn8/CQ8	DQ4
2B_T	87	IO	IOB			DIFF_IO_2B_T5n	Yes	CC31	DQ17	DQ8	DQ4
2B_T	86	IO	IOB			DIFF_IO_2B_T5p	Yes	CA31	DQ17	DQ8	DQ4
2B_T	85	IO	IOB			DIFF_IO_2B_T6n	No	CF31	DQ17	DQ8	DQ4
2B_T	84	IO	IOB			DIFF_IO_2B_T6p	No	CH31	DQ17	DQ8	DQ4
2B_T	83	IO	IOB			DIFF_IO_2B_T7n	No	CL6	DQ18	DQ9	DQ4
2B_T	82	IO	IOB			DIFF_IO_2B_T7p	No	CK8	DQ18	DQ9	DQ4
2B_T	81	IO	IOB			DIFF_IO_2B_T8n	No	CL8	DQ18	DQ9	DQ4
2B_T	80	IO	IOB			DIFF_IO_2B_T8p	No	CK11	DQ18	DQ9	DQ4
2B_T	79	IO	IOB			DIFF_IO_2B_T9n	No	CL11	DQSn18	DQ9	DQ4
2B_T	78	IO	IOB			DIFF_IO_2B_T9p	No	CL14	DQS18	DQ9	DQ4
2B_T	77	IO	IOB			DIFF_IO_2B_T10n	No	CL17	DQSn19	DQSn9/CQn9	DQSn4/CQn4
2B_T	76	IO	IOB			DIFF_IO_2B_T10p	No	CK17	DQS19	DQSn9/CQ9	DQSn4/CQ4
2B_T	75	IO	IOB			DIFF_IO_2B_T11n	Yes	CK20	DQ19	DQ9	DQ4
2B_T	74	IO	IOB			DIFF_IO_2B_T11p	Yes	CL20	DQ19	DQ9	DQ4
2B_T	73	IO	IOB			DIFF_IO_2B_T12n	No	CK26	DQ19	DQ9	DQ4
2B_T	72	IO	IOB			DIFF_IO_2B_T12p	No	CL23	DQ19	DQ9	DQ4
2B_T	71	IO	IOB			DIFF_IO_2B_T13n	No	BF46	DQ20	DQ10	DQ5
2B_T	70	IO	IOB			DIFF_IO_2B_T13p	No	BE46	DQ20	DQ10	DQ5
2B_T	69	IO	IOB			DIFF_IO_2B_T14n	No	BE50	DQ20	DQ10	DQ5
2B_T	68	IO	IOB			DIFF_IO_2B_T14p	No	BF50	DQ20	DQ10	DQ5
2B_T	67	IO	IOB			DIFF_IO_2B_T15n	No	BF53	DQSn20	DQ10	DQ5
2B_T	66	IO	IOB			DIFF_IO_2B_T15p	No	BF57	DQS20	DQ10	DQ5
2B_T	65	IO	IOB,PLL_2B_T_CLKOUT1n			DIFF_IO_2B_T16n	No	BE57	DQSn21	DQSn10/CQn10	DQ5
2B_T	64	IO	IOB,PLL_2B_T_CLKOUT1p,PLL_2B_T_CLKOUT1,PLL_2B_T_FB1			DIFF_IO_2B_T16p	No	BE61	DQS21	DQSn10/CQ10	DQ5
2B_T	63	IO	IOB			DIFF_IO_2B_T17n	Yes	BF64	DQ21	DQ10	DQ5
2B_T	62	IO	IOB,RZQ_T_2B			DIFF_IO_2B_T17p	Yes	BE64	DQ21	DQ10	DQ5
2B_T	61	IO	IOB,CLK_T_2B_1n			DIFF_IO_2B_T18n	No	BE68	DQ21	DQ10	DQ5
2B_T	60	IO	IOB,CLK_T_2B_1p			DIFF_IO_2B_T18p	No	BF68	DQ21	DQ10	DQ5
2B_T	59	IO	IOB,CLK_T_2B_0n			DIFF_IO_2B_T19n	No	BM38	DQ22	DQ11	DQ5
2B_T	58	IO	IOB,CLK_T_2B_0p			DIFF_IO_2B_T19p	No	BM38	DQ22	DQ11	DQ5
2B_T	57	IO	IOB			DIFF_IO_2B_T20n	No	BM44	DQ22	DQ11	DQ5
2B_T	56	IO	IOB			DIFF_IO_2B_T20p	No	BM38	DQ22	DQ11	DQ5
2B_T	55	IO	IOB,PLL_2B_T_CLKOUT0n			DIFF_IO_2B_T21n	No	BM41	DQSn22	DQ11	DQ5
2B_T	54	IO	IOB,PLL_2B_T_CLKOUT0p,PLL_2B_T_CLKOUT0,PLL_2B_T_FB0			DIFF_IO_2B_T21p	No	BP41	DQS22	DQ11	DQ5
2B_T	53	IO	IOB			DIFF_IO_2B_T22n	No	BM49	DQSn23	DQSn11/CQn11	DQSn5/CQn5
2B_T	52	IO	IOB			DIFF_IO_2B_T22p	No	BK49	DQS23	DQSn11/CQ11	DQSn5/CQ5
2B_T	51	IO	IOB			DIFF_IO_2B_T23n	Yes	BH52	DQ23	DQ11	DQ5
2B_T	50	IO	IOB			DIFF_IO_2B_T23p	Yes	BH49	DQ23	DQ11	DQ5
2B_T	49	IO	IOB			DIFF_IO_2B_T24n	No	BP52	DQ23	DQ11	DQ5
2B_T	48	IO	IOB			DIFF_IO_2B_T24p	No	BM52	DQ23	DQ11	DQ5
2B_B	47	IO	IOB			DIFF_IO_2B_B1n	No	BW38	DQ24	DQ12	DQ6
2B_B	46	IO	IOB			DIFF_IO_2B_B1p	No	CA38	DQ24	DQ12	DQ6
2B_B	45	IO	IOB			DIFF_IO_2B_B2n	No	BU38	DQ24	DQ12	DQ6
2B_B	44	IO	IOB			DIFF_IO_2B_B2p	No	BR38	DQ24	DQ12	DQ6
2B_B	43	IO	IOB			DIFF_IO_2B_B3n	No	BU41	DQSn24	DQ12	DQ6
2B_B	42	IO	IOB			DIFF_IO_2B_B3p	No	BR41	DQS24	DQ12	DQ6
2B_B	41	IO	IOB,PLL_2B_B_CLKOUT1n			DIFF_IO_2B_B4n	No	CA49	DQSn25	DQSn12/CQn12	DQ6
2B_B	40	IO	IOB,PLL_2B_B_CLKOUT1p,PLL_2B_B_CLKOUT1,PLL_2B_B_FB1			DIFF_IO_2B_B4p	No	BW49	DQS25	DQSn12/CQ12	DQ6
2B_B	39	IO	IOB			DIFF_IO_2B_B5n	Yes	BU52	DQ25	DQ12	DQ6
2B_B	38	IO	IOB,RZQ_B_2B			DIFF_IO_2B_B5p	Yes	BR52	DQ25	DQ12	DQ6
2B_B	37	IO	IOB,CLK_B_2B_1n			DIFF_IO_2B_B6n	Yes	BU49	DQ25	DQ12	DQ6
2B_B	36	IO	IOB,CLK_B_2B_1p			DIFF_IO_2B_B6p	Yes	BR49	DQ25	DQ12	DQ6
2B_B	35	IO	IOB,CLK_B_2B_0n			DIFF_IO_2B_B7n	No	CF38	DQ26	DQ13	DQ6
2B_B	34	IO	IOB,CLK_B_2B_0p			DIFF_IO_2B_B7p	No	CH38	DQ26	DQ13	DQ6
2B_B	33	IO	IOB			DIFF_IO_2B_B8n	No	CA41	DQ26	DQ13	DQ6
2B_B	32	IO	IOB			DIFF_IO_2B_B8p	No	CC41	DQ26	DQ13	DQ6
2B_B	31	IO	IOB,PLL_2B_B_CLKOUT0n			DIFF_IO_2B_B9n	No	CF41	DQSn26	DQ13	DQ6
2B_B	30	IO	IOB,PLL_2B_B_CLKOUT0p,PLL_2B_B_CLKOUT0,PLL_2B_B_FB0			DIFF_IO_2B_B9p	No	CH41	DQS26	DQ13	DQ6
2B_B	29	IO	IOB			DIFF_IO_2B_B10n	No	CH49	DQSn27	DQSn13/CQn13	DQSn6/CQn6
2B_B	28	IO	IOB			DIFF_IO_2B_B10p	No	CF49	DQS27	DQSn13/CQ13	DQSn6/CQ6
2B_B	27	IO	IOB			DIFF_IO_2B_B11n	Yes	CA52	DQ27	DQ13	DQ6

Pin Information for Aglex™ 5 A5ED052A Device											
Version: 2025-06-06											
B32A Status: Final											
Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
2B_B	26	IO	IOB			DIFF_IO_2B_B11p	Yes	CC52	DQ27	DQ13	DQ6
2B_B	25	IO	IOB			DIFF_IO_2B_B12n	Yes	CH52	DQ27	DQ13	DQ6
2B_B	24	IO	IOB			DIFF_IO_2B_B12p	Yes	CF52	DQ27	DQ13	DQ6
2B_B	23	IO	IOB			DIFF_IO_2B_B13n	No	CL26	DQ28	DQ14	DQ7
2B_B	22	IO	IOB			DIFF_IO_2B_B13p	No	CK30	DQ28	DQ14	DQ7
2B_B	21	IO	IOB			DIFF_IO_2B_B14n	No	CL30	DQ28	DQ14	DQ7
2B_B	20	IO	IOB			DIFF_IO_2B_B14p	No	CK33	DQ28	DQ14	DQ7
2B_B	19	IO	IOB			DIFF_IO_2B_B15n	No	CL35	DQSn28	DQ14	DQ7
2B_B	18	IO	IOB			DIFF_IO_2B_B15p	No	CK35	DQS28	DQ14	DQ7
2B_B	17	IO	IOB			DIFF_IO_2B_B16n	No	CL39	DQSn29	DQSn14/CQn14	DQ7
2B_B	16	IO	IOB			DIFF_IO_2B_B16p	No	CK39	DQS29	DQS14/CQ14	DQ7
2B_B	15	IO	IOB			DIFF_IO_2B_B17n	Yes	CL45	DQ29	DQ14	DQ7
2B_B	14	IO	IOB			DIFF_IO_2B_B17p	Yes	CK48	DQ29	DQ14	DQ7
2B_B	13	IO	IOB			DIFF_IO_2B_B18n	Yes	CK45	DQ29	DQ14	DQ7
2B_B	12	IO	IOB			DIFF_IO_2B_B18p	Yes	CL42	DQ29	DQ14	DQ7
2B_B	11	IO	IOB			DIFF_IO_2B_B19n	No	CK54	DQ30	DQ15	DQ7
2B_B	10	IO	IOB			DIFF_IO_2B_B19p	No	CL51	DQ30	DQ15	DQ7
2B_B	9	IO	IOB			DIFF_IO_2B_B20n	No	CL54	DQ30	DQ15	DQ7
2B_B	8	IO	IOB			DIFF_IO_2B_B20p	No	CK56	DQ30	DQ15	DQ7
2B_B	7	IO	IOB			DIFF_IO_2B_B21n	No	CL60	DQSn30	DQ15	DQ7
2B_B	6	IO	IOB			DIFF_IO_2B_B21p	No	CL56	DQS30	DQ15	DQ7
2B_B	5	IO	IOB			DIFF_IO_2B_B22n	No	CL66	DQSn31	DQSn15/CQn15	DQSn7/CQn7
2B_B	4	IO	IOB			DIFF_IO_2B_B22p	No	CK63	DQSn31	DQS15/CQ15	DQSn7/CQ7
2B_B	3	IO	IOB			DIFF_IO_2B_B23n	Yes	CL73	DQ31	DQ15	DQ7
2B_B	2	IO	IOB			DIFF_IO_2B_B23p	Yes	CK73	DQ31	DQ15	DQ7
2B_B	1	IO	IOB			DIFF_IO_2B_B24n	Yes	CL70	DQ31	DQ15	DQ7
2B_B	0	IO	IOB			DIFF_IO_2B_B24p	Yes	CK66	DQ31	DQ15	DQ7
3A_T	95	IO	IOB		HPS_DDR	DIFF_IO_3A_T1n	No	A91	DQ32	DQ16	DQ8
3A_T	94	IO	IOB		HPS_DDR	DIFF_IO_3A_T1p	No	B88	DQ32	DQ16	DQ8
3A_T	93	IO	IOB		HPS_DDR	DIFF_IO_3A_T2n	No	A94	DQ32	DQ16	DQ8
3A_T	92	IO	IOB		HPS_DDR	DIFF_IO_3A_T2p	No	B91	DQ32	DQ16	DQ8
3A_T	91	IO	IOB		HPS_DDR	DIFF_IO_3A_T3n	No	A97	DQSn32	DQ16	DQ8
3A_T	90	IO	IOB	AVST_READY	HPS_DDR	DIFF_IO_3A_T3p	No	B97	DQS32	DQ16	DQ8
3A_T	89	IO	IOB		HPS_DDR	DIFF_IO_3A_T4n	No	B101	DQSn33	DQSn16/CQn16	DQ8
3A_T	88	IO	IOB		HPS_DDR	DIFF_IO_3A_T4p	No	A101	DQSn33	DQS16/CQ16	DQ8
3A_T	87	IO	IOB		HPS_DDR	DIFF_IO_3A_T5n	Yes	A110	DQ33	DQ16	DQ8
3A_T	86	IO	IOB		HPS_DDR	DIFF_IO_3A_T5p	Yes	B106	DQ33	DQ16	DQ8
3A_T	85	IO	IOB		HPS_DDR	DIFF_IO_3A_T6n	No	B103	DQ33	DQ16	DQ8
3A_T	84	IO	IOB		HPS_DDR	DIFF_IO_3A_T6p	No	A106	DQ33	DQ16	DQ8
3A_T	83	IO	IOB	AVST_DATA10	HPS_DDR	DIFF_IO_3A_T7n	No	D84	DQ34	DQ17	DQ8
3A_T	82	IO	IOB	AVST_DATA9	HPS_DDR	DIFF_IO_3A_T7p	No	F84	DQ34	DQ17	DQ8
3A_T	81	IO	IOB	AVST_DATA8	HPS_DDR	DIFF_IO_3A_T8n	No	M87	DQ34	DQ17	DQ8
3A_T	80	IO	IOB	AVST_VALID	HPS_DDR	DIFF_IO_3A_T8p	No	K87	DQ34	DQ17	DQ8
3A_T	79	IO	IOB	AVST_DATA7	HPS_DDR	DIFF_IO_3A_T9n	No	F87	DQSn34	DQ17	DQ8
3A_T	78	IO	IOB	AVST_DATA6	HPS_DDR	DIFF_IO_3A_T9p	No	H87	DQ34	DQ17	DQ8
3A_T	77	IO	IOB	AVST_DATA5	HPS_DDR	DIFF_IO_3A_T10n	No	D95	DQSn35	DQSn17/CQn17	DQSn8/CQn8
3A_T	76	IO	IOB	AVST_DATA4	HPS_DDR	DIFF_IO_3A_T10p	No	F95	DQSn35	DQS17/CQ17	DQSn8/CQ8
3A_T	75	IO	IOB	AVST_DATA3	HPS_DDR	DIFF_IO_3A_T11n	Yes	K98	DQ35	DQ17	DQ8
3A_T	74	IO	IOB	AVST_DATA2	HPS_DDR	DIFF_IO_3A_T11p	Yes	M98	DQ35	DQ17	DQ8
3A_T	73	IO	IOB	AVST_DATA1	HPS_DDR	DIFF_IO_3A_T12n	No	F98	DQ35	DQ17	DQ8
3A_T	72	IO	IOB	AVST_DATA0	HPS_DDR	DIFF_IO_3A_T12p	No	H98	DQ35	DQ17	DQ8
3A_T	71	IO	IOB		HPS_DDR	DIFF_IO_3A_T13n	No	P84	DQ36	DQ18	DQ9
3A_T	70	IO	IOB		HPS_DDR	DIFF_IO_3A_T13p	No	T84	DQ36	DQ18	DQ9
3A_T	69	IO	IOB		HPS_DDR	DIFF_IO_3A_T14n	No	M84	DQ36	DQ18	DQ9
3A_T	68	IO	IOB		HPS_DDR	DIFF_IO_3A_T14p	No	K84	DQ36	DQ18	DQ9
3A_T	67	IO	IOB		HPS_DDR	DIFF_IO_3A_T15n	No	T87	DQSn36	DQ18	DQ9
3A_T	66	IO	IOB		HPS_DDR	DIFF_IO_3A_T15p	No	V87	DQ36	DQ18	DQ9
3A_T	65	IO	IOB,PLL_3A_T_CLKOUT1n		HPS_DDR	DIFF_IO_3A_T16n	No	M95	DQSn37	DQSn18/CQn18	DQ9
3A_T	64	IO	IOB,PLL_3A_T_CLKOUT1p,PLL_3A_T_CLKOUT1,PLL_3A_T_FB1		HPS_DDR	DIFF_IO_3A_T16p	No	K95	DQSn37	DQS18/CQ18	DQ9
3A_T	63	IO	IOB		HPS_DDR	DIFF_IO_3A_T17n	Yes	T95	DQ37	DQ18	DQ9
3A_T	62	IO	IOB,RZQ_T_3A		HPS_DDR	DIFF_IO_3A_T17p	Yes	P95	DQ37	DQ18	DQ9
3A_T	61	IO	IOB,CLK_T_3A_1n		HPS_DDR	DIFF_IO_3A_T18n	No	T98	DQ37	DQ18	DQ9
3A_T	60	IO	IOB,CLK_T_3A_1p		HPS_DDR	DIFF_IO_3A_T18p	No	V98	DQ37	DQ18	DQ9
3A_T	59	IO	IOB,CLK_T_3A_0n		HPS_DDR	DIFF_IO_3A_T19n	No	Y84	DQ38	DQ19	DQ9
3A_T	58	IO	IOB,CLK_T_3A_0p		HPS_DDR	DIFF_IO_3A_T19p	No	Y87	DQ38	DQ19	DQ9
3A_T	57	IO	IOB		HPS_DDR	DIFF_IO_3A_T20n	No	Y95	DQ38	DQ19	DQ9
3A_T	56	IO	IOB		HPS_DDR	DIFF_IO_3A_T20p	No	Y98	DQ38	DQ19	DQ9
3A_T	55	IO	IOB,PLL_3A_T_CLKOUT0n		HPS_DDR	DIFF_IO_3A_T21n	No	AC96	DQSn38	DQ19	DQ9
3A_T	54	IO	IOB,PLL_3A_T_CLKOUT0p,PLL_3A_T_CLKOUT0,PLL_3A_T_FB0		HPS_DDR	DIFF_IO_3A_T21p	No	AC90	DQSn38	DQ19	DQ9
3A_T	53	IO	IOB	AVST_CLK	HPS_DDR	DIFF_IO_3A_T22n	No	AG93	DQSn39	DQSn19/CQn19	DQSn9/CQn9
3A_T	52	IO	IOB	AVST_DATA15	HPS_DDR	DIFF_IO_3A_T22p	No	AG90	DQSn39	DQS19/CQ19	DQSn9/CQ9
3A_T	51	IO	IOB	AVST_DATA14	HPS_DDR	DIFF_IO_3A_T23n	Yes	AC96	DQ39	DQ19	DQ9
3A_T	50	IO	IOB	AVST_DATA13	HPS_DDR	DIFF_IO_3A_T23p	Yes	AC100	DQ39	DQ19	DQ9
3A_T	49	IO	IOB	AVST_DATA12	HPS_DDR	DIFF_IO_3A_T24n	No	AG104	DQ39	DQ19	DQ9
3A_T	48	IO	IOB	AVST_DATA11	HPS_DDR	DIFF_IO_3A_T24p	No	AG100	DQ39	DQ19	DQ9
3A_B	47	IO	IOB		HPS_DDR	DIFF_IO_3A_B1n	No	AB105	DQ40	DQ20	DQ10
3A_B	46	IO	IOB		HPS_DDR	DIFF_IO_3A_B1p	No	Y105	DQ40	DQ20	DQ10
3A_B	45	IO	IOB		HPS_DDR	DIFF_IO_3A_B2n	No	AB108	DQ40	DQ20	DQ10
3A_B	44	IO	IOB		HPS_DDR	DIFF_IO_3A_B2p	No	Y108	DQ40	DQ20	DQ10
3A_B	43	IO	IOB		HPS_DDR	DIFF_IO_3A_B3n	No	AK104	DQSn40	DQ20	DQ10
3A_B	42	IO	IOB		HPS_DDR	DIFF_IO_3A_B3p	No	AK107	DQSn40	DQ20	DQ10
3A_B	41	IO	IOB,PLL_3A_B_CLKOUT1n		HPS_DDR	DIFF_IO_3A_B4n	No	AB114	DQSn41	DQSn20/CQn20	DQ10

altera				Pin Information for Aglex™ 5 A5ED052A Device							
				Version: 2025-06-06							
				B32A Status: Final							
Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
3A, B	40	IO	IOB,PLL_3A_B_CLKOUT1p,PLL_3A_B_CLKOUT1,PLL_3A_B_FB1		HPS_DDR	DIFF_IO_3A_B4p	No	V114	DQS41	DQS20/CQ20	DQ10
3A, B	39	IO	IOB		HPS_DDR	DIFF_IO_3A_B5n	Yes	AG111	DQ41	DQ20	DQ10
3A, B	38	IO	IOB,RZQ_B_3A		HPS_DDR	DIFF_IO_3A_B5p	Yes	AK111	DQ41	DQ20	DQ10
3A, B	37	IO	IOB,CLK_B_3A_1n		HPS_DDR	DIFF_IO_3A_B6n	Yes	Y117	DQ41	DQ20	DQ10
3A, B	36	IO	IOB,CLK_B_3A_1p		HPS_DDR	DIFF_IO_3A_B6p	Yes	AB117	DQ41	DQ20	DQ10
3A, B	35	IO	IOB,CLK_B_3A_0n		HPS_DDR	DIFF_IO_3A_B7n	No	K105	DQ42	DQ21	DQ10
3A, B	34	IO	IOB,CLK_B_3A_0p		HPS_DDR	DIFF_IO_3A_B7p	No	M105	DQ42	DQ21	DQ10
3A, B	33	IO	IOB		HPS_DDR	DIFF_IO_3A_B8n	No	P105	DQ42	DQ21	DQ10
3A, B	32	IO	IOB		HPS_DDR	DIFF_IO_3A_B8p	No	T105	DQ42	DQ21	DQ10
3A, B	31	IO	IOB,PLL_3A_B_CLKOUT0n		HPS_DDR	DIFF_IO_3A_B9n	No	T108	DQSn42	DQ21	DQ10
3A, B	30	IO	IOB,PLL_3A_B_CLKOUT0p,PLL_3A_B_CLKOUT0,PLL_3A_B_FB0		HPS_DDR	DIFF_IO_3A_B9p	No	V108	DQS42	DQ21	DQ10
3A, B	29	IO	IOB		HPS_DDR	DIFF_IO_3A_B10n	No	K114	DQSn43	DQSn21/CQn21	DQSn10/CQn10
3A, B	28	IO	IOB		HPS_DDR	DIFF_IO_3A_B10p	No	M114	DQS43	DQS21/CQ21	DQSn10/CQ10
3A, B	27	IO	IOB		HPS_DDR	DIFF_IO_3A_B11n	Yes	T117	DQ43	DQ21	DQ10
3A, B	26	IO	IOB		HPS_DDR	DIFF_IO_3A_B11p	Yes	V117	DQ43	DQ21	DQ10
3A, B	25	IO	IOB		HPS_DDR	DIFF_IO_3A_B12n	Yes	P114	DQ43	DQ21	DQ10
3A, B	24	IO	IOB		HPS_DDR	DIFF_IO_3A_B12p	Yes	T114	DQ43	DQ21	DQ10
3A, B	23	IO	IOB		HPS_DDR	DIFF_IO_3A_B13n	No	K108	DQ44	DQ22	DQ11
3A, B	22	IO	IOB		HPS_DDR	DIFF_IO_3A_B13p	No	M108	DQ44	DQ22	DQ11
3A, B	21	IO	IOB		HPS_DDR	DIFF_IO_3A_B14n	No	F108	DQ44	DQ22	DQ11
3A, B	20	IO	IOB		HPS_DDR	DIFF_IO_3A_B14p	No	H108	DQ44	DQ22	DQ11
3A, B	19	IO	IOB		HPS_DDR	DIFF_IO_3A_B15n	No	D105	DQSn44	DQ22	DQ11
3A, B	18	IO	IOB		HPS_DDR	DIFF_IO_3A_B15p	No	F105	DQS44	DQ22	DQ11
3A, B	17	IO	IOB		HPS_DDR	DIFF_IO_3A_B16n	No	D114	DQSn45	DQSn22/CQn22	DQ11
3A, B	16	IO	IOB		HPS_DDR	DIFF_IO_3A_B16p	No	F114	DQS45	DQS22/CQ22	DQ11
3A, B	15	IO	IOB		HPS_DDR	DIFF_IO_3A_B17n	Yes	M117	DQ45	DQ22	DQ11
3A, B	14	IO	IOB		HPS_DDR	DIFF_IO_3A_B17p	Yes	K117	DQ45	DQ22	DQ11
3A, B	13	IO	IOB		HPS_DDR	DIFF_IO_3A_B18n	Yes	H117	DQ45	DQ22	DQ11
3A, B	12	IO	IOB		HPS_DDR	DIFF_IO_3A_B18p	Yes	F117	DQ45	DQ22	DQ11
3A, B	11	IO	IOB		HPS_DDR	DIFF_IO_3A_B19n	No	A113	DQ46	DQ23	DQ11
3A, B	10	IO	IOB		HPS_DDR	DIFF_IO_3A_B19p	No	B113	DQ46	DQ23	DQ11
3A, B	9	IO	IOB		HPS_DDR	DIFF_IO_3A_B20n	No	A116	DQ46	DQ23	DQ11
3A, B	8	IO	IOB		HPS_DDR	DIFF_IO_3A_B20p	No	B116	DQ46	DQ23	DQ11
3A, B	7	IO	IOB		HPS_DDR	DIFF_IO_3A_B21n	No	A122	DQSn46	DQ23	DQ11
3A, B	6	IO	IOB		HPS_DDR	DIFF_IO_3A_B21p	No	B119	DQS46	DQ23	DQ11
3A, B	5	IO	IOB		HPS_DDR	DIFF_IO_3A_B22n	No	A125	DQSn47	DQSn23/CQn23	DQSn11/CQn11
3A, B	4	IO	IOB		HPS_DDR	DIFF_IO_3A_B22p	No	B122	DQS47	DQS23/CQ23	DQSn11/CQ11
3A, B	3	IO	IOB		HPS_DDR	DIFF_IO_3A_B23n	Yes	A130	DQ47	DQ23	DQ11
3A, B	2	IO	IOB		HPS_DDR	DIFF_IO_3A_B23p	Yes	B130	DQ47	DQ23	DQ11
3A, B	1	IO	IOB		HPS_DDR	DIFF_IO_3A_B24n	Yes	A128	DQ47	DQ23	DQ11
3A, B	0	IO	IOB		HPS_DDR	DIFF_IO_3A_B24p	Yes	B128	DQ47	DQ23	DQ11
3B, T	95	IO	IOB		HPS_DDR	DIFF_IO_3B_T1n	No	B42	DQ48	DQ24	DQ12
3B, T	94	IO	IOB		HPS_DDR	DIFF_IO_3B_T1p	No	A45	DQ48	DQ24	DQ12
3B, T	93	IO	IOB		HPS_DDR	DIFF_IO_3B_T2n	No	A48	DQ48	DQ24	DQ12
3B, T	92	IO	IOB		HPS_DDR	DIFF_IO_3B_T2p	No	B45	DQ48	DQ24	DQ12
3B, T	91	IO	IOB		HPS_DDR	DIFF_IO_3B_T3n	No	A51	DQSn48	DQ24	DQ12
3B, T	90	IO	IOB		HPS_DDR	DIFF_IO_3B_T3p	No	B51	DQS48	DQ24	DQ12
3B, T	89	IO	IOB		HPS_DDR	DIFF_IO_3B_T4n	No	B54	DQSn49	DQSn24/CQn24	DQ12
3B, T	88	IO	IOB		HPS_DDR	DIFF_IO_3B_T4p	No	A54	DQS49	DQS24/CQ24	DQ12
3B, T	87	IO	IOB		HPS_DDR	DIFF_IO_3B_T5n	Yes	B60	DQ49	DQ24	DQ12
3B, T	86	IO	IOB		HPS_DDR	DIFF_IO_3B_T5p	Yes	A63	DQ49	DQ24	DQ12
3B, T	85	IO	IOB		HPS_DDR	DIFF_IO_3B_T6n	No	A60	DQ49	DQ24	DQ12
3B, T	84	IO	IOB		HPS_DDR	DIFF_IO_3B_T6p	No	B56	DQ49	DQ24	DQ12
3B, T	83	IO	IOB		HPS_DDR	DIFF_IO_3B_T7n	No	F44	DQ50	DQ25	DQ12
3B, T	82	IO	IOB		HPS_DDR	DIFF_IO_3B_T7p	No	D44	DQ50	DQ25	DQ12
3B, T	81	IO	IOB		HPS_DDR	DIFF_IO_3B_T8n	No	H47	DQ50	DQ25	DQ12
3B, T	80	IO	IOB		HPS_DDR	DIFF_IO_3B_T8p	No	F47	DQ50	DQ25	DQ12
3B, T	79	IO	IOB		HPS_DDR	DIFF_IO_3B_T9n	No	K47	DQSn50	DQ25	DQ12
3B, T	78	IO	IOB		HPS_DDR	DIFF_IO_3B_T9p	No	M47	DQSn50	DQ25	DQ12
3B, T	77	IO	IOB		HPS_DDR	DIFF_IO_3B_T10n	No	D55	DQSn51	DQSn25/CQn25	DQSn12/CQn12
3B, T	76	IO	IOB		HPS_DDR	DIFF_IO_3B_T10p	No	F55	DQSn51	DQS25/CQ25	DQSn12/CQ12
3B, T	75	IO	IOB		HPS_DDR	DIFF_IO_3B_T11n	Yes	K58	DQ51	DQ25	DQ12
3B, T	74	IO	IOB		HPS_DDR	DIFF_IO_3B_T11p	Yes	M58	DQ51	DQ25	DQ12
3B, T	73	IO	IOB		HPS_DDR	DIFF_IO_3B_T12n	No	F58	DQ51	DQ25	DQ12
3B, T	72	IO	IOB		HPS_DDR	DIFF_IO_3B_T12p	No	H58	DQ51	DQ25	DQ12
3B, T	71	IO	IOB		HPS_DDR	DIFF_IO_3B_T13n	No	M44	DQ52	DQ26	DQ13
3B, T	70	IO	IOB		HPS_DDR	DIFF_IO_3B_T13p	No	K44	DQ52	DQ26	DQ13
3B, T	69	IO	IOB		HPS_DDR	DIFF_IO_3B_T14n	No	T44	DQ52	DQ26	DQ13
3B, T	68	IO	IOB		HPS_DDR	DIFF_IO_3B_T14p	No	P44	DQ52	DQ26	DQ13
3B, T	67	IO	IOB		HPS_DDR	DIFF_IO_3B_T15n	No	T47	DQSn52	DQ26	DQ13
3B, T	66	IO	IOB		HPS_DDR	DIFF_IO_3B_T15p	No	V47	DQSn52	DQ26	DQ13
3B, T	65	IO	IOB,PLL_3B_T_CLKOUT1n		HPS_DDR	DIFF_IO_3B_T16n	No	M55	DQSn53	DQSn26/CQn26	DQ13
3B, T	64	IO	IOB,PLL_3B_T_CLKOUT1p,PLL_3B_T_CLKOUT1,PLL_3B_T_FB1		HPS_DDR	DIFF_IO_3B_T16p	No	K55	DQSn53	DQS26/CQ26	DQ13
3B, T	63	IO	IOB		HPS_DDR	DIFF_IO_3B_T17n	Yes	T58	DQ53	DQ26	DQ13
3B, T	62	IO	IOB,RZQ_T_3B		HPS_DDR	DIFF_IO_3B_T17p	Yes	V58	DQ53	DQ26	DQ13
3B, T	61	IO	IOB,CLK_T_3B_1n		HPS_DDR	DIFF_IO_3B_T18n	No	T55	DQ53	DQ26	DQ13
3B, T	60	IO	IOB,CLK_T_3B_1p		HPS_DDR	DIFF_IO_3B_T18p	No	P55	DQ53	DQ26	DQ13
3B, T	59	IO	IOB,CLK_T_3B_0n		HPS_DDR	DIFF_IO_3B_T19n	No	Y47	DQ54	DQ27	DQ13
3B, T	58	IO	IOB,CLK_T_3B_0p		HPS_DDR	DIFF_IO_3B_T19p	No	Y44	DQ54	DQ27	DQ13
3B, T	57	IO	IOB		HPS_DDR	DIFF_IO_3B_T20n	No	Y55	DQ54	DQ27	DQ13
3B, T	56	IO	IOB		HPS_DDR	DIFF_IO_3B_T20p	No	Y58	DQ54	DQ27	DQ13
3B, T	55	IO	IOB,PLL_3B_T_CLKOUT0n		HPS_DDR	DIFF_IO_3B_T21n	No	AC50	DQSn54	DQ27	DQ13

altera				Pin Information for Aglex™ 5 A5ED052A Device							
				Version: 2025-06-06							
				B32A Status: Final							
Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
3B_T	54	IO	IOB,PLL_3B_T_CLKOUT0p,PLL_3B_T_CLKOUT0,PLL_3B_T_FB0		HPS_DDR	DIFF_IO_3B_T21p	No	AC53	DQ554	DQ27	DQ13
3B_T	53	IO	IOB		HPS_DDR	DIFF_IO_3B_T22m	No	AG53	DQ5n55	DQ5n27/CQn27	DQ5n13/CQn13
3B_T	52	IO	IOB		HPS_DDR	DIFF_IO_3B_T22p	No	AG57	DQ555	DQ527/CQ27	DQ513/CQ13
3B_T	51	IO	IOB		HPS_DDR	DIFF_IO_3B_T23n	Yes	AC61	DQ55	DQ27	DQ13
3B_T	50	IO	IOB		HPS_DDR	DIFF_IO_3B_T23p	Yes	AG61	DQ55	DQ27	DQ13
3B_T	49	IO	IOB		HPS_DDR	DIFF_IO_3B_T24n	No	AG64	DQ55	DQ27	DQ13
3B_T	48	IO	IOB		HPS_DDR	DIFF_IO_3B_T24p	No	AC64	DQ55	DQ27	DQ13
3B_B	47	IO	IOB		HPS_DDR	DIFF_IO_3B_B1n	No	Y65	DQ56	DQ28	DQ14
3B_B	46	IO	IOB		HPS_DDR	DIFF_IO_3B_B1p	No	Y67	DQ56	DQ28	DQ14
3B_B	45	IO	IOB		HPS_DDR	DIFF_IO_3B_B2n	No	Y74	DQ56	DQ28	DQ14
3B_B	44	IO	IOB		HPS_DDR	DIFF_IO_3B_B2p	No	Y77	DQ56	DQ28	DQ14
3B_B	43	IO	IOB		HPS_DDR	DIFF_IO_3B_B3n	No	AC83	DQ5n56	DQ28	DQ14
3B_B	42	IO	IOB		HPS_DDR	DIFF_IO_3B_B3p	No	AG83	DQ556	DQ28	DQ14
3B_B	41	IO	IOB,PLL_3B_B_CLKOUT1n		HPS_DDR	DIFF_IO_3B_B4n	No	AG75	DQ5n57	DQ5n28/CQn28	DQ14
3B_B	40	IO	IOB,PLL_3B_B_CLKOUT1p,PLL_3B_B_CLKOUT1,PLL_3B_B_FB1		HPS_DDR	DIFF_IO_3B_B4p	No	AG72	DQ557	DQ528/CQ28	DQ14
3B_B	39	IO	IOB		HPS_DDR	DIFF_IO_3B_B5n	Yes	AG79	DQ57	DQ28	DQ14
3B_B	38	IO	IOB,RZQ_B_3B		HPS_DDR	DIFF_IO_3B_B5p	Yes	AC79	DQ57	DQ28	DQ14
3B_B	37	IO	IOB,CLK_B_3B_1n		HPS_DDR	DIFF_IO_3B_B6n	Yes	AC72	DQ57	DQ28	DQ14
3B_B	36	IO	IOB,CLK_B_3B_1p		HPS_DDR	DIFF_IO_3B_B6p	Yes	AC68	DQ57	DQ28	DQ14
3B_B	35	IO	IOB,CLK_B_3B_0n		HPS_DDR	DIFF_IO_3B_B7n	No	P65	DQ58	DQ29	DQ14
3B_B	34	IO	IOB,CLK_B_3B_0p		HPS_DDR	DIFF_IO_3B_B7p	No	T65	DQ58	DQ29	DQ14
3B_B	33	IO	IOB		HPS_DDR	DIFF_IO_3B_B8n	No	K65	DQ58	DQ29	DQ14
3B_B	32	IO	IOB		HPS_DDR	DIFF_IO_3B_B8p	No	M65	DQ58	DQ29	DQ14
3B_B	31	IO	IOB,PLL_3B_B_CLKOUT0n		HPS_DDR	DIFF_IO_3B_B9n	No	T67	DQ5n58	DQ29	DQ14
3B_B	30	IO	IOB,PLL_3B_B_CLKOUT0p,PLL_3B_B_CLKOUT0,PLL_3B_B_FB0		HPS_DDR	DIFF_IO_3B_B9p	No	V67	DQ558	DQ29	DQ14
3B_B	29	IO	IOB		HPS_DDR	DIFF_IO_3B_B10n	No	K74	DQ5n59	DQ5n29/CQn29	DQ5n14/CQn14
3B_B	28	IO	IOB		HPS_DDR	DIFF_IO_3B_B10p	No	M74	DQ559	DQ529/CQ29	DQ514/CQ14
3B_B	27	IO	IOB		HPS_DDR	DIFF_IO_3B_B11n	Yes	T77	DQ59	DQ29	DQ14
3B_B	26	IO	IOB		HPS_DDR	DIFF_IO_3B_B11p	Yes	V77	DQ59	DQ29	DQ14
3B_B	25	IO	IOB		HPS_DDR	DIFF_IO_3B_B12n	Yes	T74	DQ59	DQ29	DQ14
3B_B	24	IO	IOB		HPS_DDR	DIFF_IO_3B_B12p	Yes	P74	DQ59	DQ29	DQ14
3B_B	23	IO	IOB		HPS_DDR	DIFF_IO_3B_B13n	No	D65	DQ60	DQ30	DQ15
3B_B	22	IO	IOB		HPS_DDR	DIFF_IO_3B_B13p	No	F65	DQ60	DQ30	DQ15
3B_B	21	IO	IOB		HPS_DDR	DIFF_IO_3B_B14n	No	F67	DQ60	DQ30	DQ15
3B_B	20	IO	IOB		HPS_DDR	DIFF_IO_3B_B14p	No	H67	DQ60	DQ30	DQ15
3B_B	19	IO	IOB		HPS_DDR	DIFF_IO_3B_B15n	No	K67	DQ5n60	DQ30	DQ15
3B_B	18	IO	IOB		HPS_DDR	DIFF_IO_3B_B15p	No	M67	DQ560	DQ30	DQ15
3B_B	17	IO	IOB		HPS_DDR	DIFF_IO_3B_B16n	No	F74	DQ5n61	DQ5n30/CQn30	DQ15
3B_B	16	IO	IOB		HPS_DDR	DIFF_IO_3B_B16p	No	D74	DQ561	DQ530/CQ30	DQ15
3B_B	15	IO	IOB		HPS_DDR	DIFF_IO_3B_B17n	Yes	H77	DQ61	DQ30	DQ15
3B_B	14	IO	IOB		HPS_DDR	DIFF_IO_3B_B17p	Yes	F77	DQ61	DQ30	DQ15
3B_B	13	IO	IOB		HPS_DDR	DIFF_IO_3B_B18n	Yes	M77	DQ61	DQ30	DQ15
3B_B	12	IO	IOB		HPS_DDR	DIFF_IO_3B_B18p	Yes	K77	DQ61	DQ30	DQ15
3B_B	11	IO	IOB		HPS_DDR	DIFF_IO_3B_B19n	No	A66	DQ62	DQ31	DQ15
3B_B	10	IO	IOB		HPS_DDR	DIFF_IO_3B_B19p	No	B66	DQ62	DQ31	DQ15
3B_B	9	IO	IOB		HPS_DDR	DIFF_IO_3B_B20n	No	A70	DQ62	DQ31	DQ15
3B_B	8	IO	IOB		HPS_DDR	DIFF_IO_3B_B20p	No	B70	DQ62	DQ31	DQ15
3B_B	7	IO	IOB		HPS_DDR	DIFF_IO_3B_B21n	No	A76	DQ5n62	DQ31	DQ15
3B_B	6	IO	IOB		HPS_DDR	DIFF_IO_3B_B21p	No	B73	DQ562	DQ31	DQ15
3B_B	5	IO	IOB		HPS_DDR	DIFF_IO_3B_B22n	No	B76	DQ5n63	DQ5n31/CQn31	DQ5n15/CQn15
3B_B	4	IO	IOB		HPS_DDR	DIFF_IO_3B_B22p	No	A80	DQ563	DQ531/CQ31	DQ515/CQ15
3B_B	3	IO	IOB		HPS_DDR	DIFF_IO_3B_B23n	Yes	A85	DQ63	DQ31	DQ15
3B_B	2	IO	IOB		HPS_DDR	DIFF_IO_3B_B23p	Yes	B85	DQ63	DQ31	DQ15
3B_B	1	IO	IOB		HPS_DDR	DIFF_IO_3B_B24n	Yes	B82	DQ63	DQ31	DQ15
3B_B	0	IO	IOB		HPS_DDR	DIFF_IO_3B_B24p	Yes	A82	DQ63	DQ31	DQ15
5A		IO	HVIO_5A_1,TXCLK1,Data_Ctrl1					CD134			
5A		IO	HVIO_5A_2,TXCLK2,Data_Ctrl2					CD135			
5A		IO	HVIO_5A_3,TXCLK3,Data_Ctrl3					CG134			
5A		IO	HVIO_5A_4,TXCLK4,Data_Ctrl4					CG135			
5A		IO	HVIO_5A_5,PIN_PERST_N_CVP_L1A_0,TXCLK5,Data_Ctrl5					CH132			
5A		IO	HVIO_5A_6,PIN_PERST_N_CVP_L1B_0,TXCLK6,Data_Ctrl6					CF132			
5A		IO	HVIO_5A_7,PIN_PERST_N_CVP_L1C_0,TXCLK7,Data_Ctrl7					CF128			
5A		IO	HVIO_5A_8,TXCLK8,Data_Ctrl8					CK134			
5A		IO	HVIO_5A_9,PLLREFCLK1,TXCLK9,RXCLK1,Data_Ctrl9					CH128			
5A		IO	HVIO_5A_10,PLLREFCLK2,TXCLK10,RXCLK2,Data_Ctrl10					CL125			
5A		IO	HVIO_5A_11,SOURCE_SYNC_CLK1,TXCLK11,RXCLK3,Data_Ctrl11					CF121			
5A		IO	HVIO_5A_12,SOURCE_SYNC_CLK2,TXCLK12,RXCLK4,Data_Ctrl12					CF118			
5A		IO	HVIO_5A_13,TXCLK13,Data_Ctrl13					BU118			
5A		IO	HVIO_5A_14,TXCLK14,Data_Ctrl14					BR118			
5A		IO	HVIO_5A_15,TXCLK15,Data_Ctrl15					CA118			
5A		IO	HVIO_5A_16,TXCLK16,Data_Ctrl16					BW118			
5A		IO	HVIO_5A_17,TXCLK17,Data_Ctrl17					CL128			
5A		IO	HVIO_5A_18,TXCLK18,Data_Ctrl18					CL130			
5A		IO	HVIO_5A_19,TXCLK19,Data_Ctrl19					CK125			
5A		IO	HVIO_5A_20,TXCLK20,Data_Ctrl20					CK128			
5B		IO	HVIO_5B_1,SYSPLLREFCLK_L1A_0,TXCLK1,Data_Ctrl1					BF111			
5B		IO	HVIO_5B_2,SYSPLLREFCLK_L1A_1,TXCLK2,Data_Ctrl2					BH109			
5B		IO	HVIO_5B_3,SYSPLLREFCLK_L1B_0,TXCLK3,Data_Ctrl3					BE115			
5B		IO	HVIO_5B_4,SYSPLLREFCLK_L1B_1,TXCLK4,Data_Ctrl4					BF115			
5B		IO	HVIO_5B_5,PIN_PERST_N_CVP_L1A_1,TXCLK5,Data_Ctrl5					BF107			
5B		IO	HVIO_5B_6,PIN_PERST_N_CVP_L1B_1,TXCLK6,Data_Ctrl6					BU109			
5B		IO	HVIO_5B_7,PIN_PERST_N_CVP_L1C_1,TXCLK7,Data_Ctrl7					BF104			

altera Pin Information for Aglex™ 5 A5ED052A Device Version: 2025-06-06 B32A Status: Final											
Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
5B		IO	HVIO_5B_8, TXCLK8, Data_Ctrl8					BR109			
5B		IO	HVIO_5B_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					BE107			
5B		IO	HVIO_5B_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					BK109			
5B		IO	HVIO_5B_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					BE111			
5B		IO	HVIO_5B_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					BM109			
5B		IO	HVIO_5B_13, TXCLK13, Data_Ctrl13					BR112			
5B		IO	HVIO_5B_14, TXCLK14, Data_Ctrl14					BK118			
5B		IO	HVIO_5B_15, TXCLK15, Data_Ctrl15					BM118			
5B		IO	HVIO_5B_16, TXCLK16, Data_Ctrl16					BP112			
5B		IO	HVIO_5B_17, TXCLK17, Data_Ctrl17					BM112			
5B		IO	HVIO_5B_18, TXCLK18, Data_Ctrl18					BK112			
5B		IO	HVIO_5B_19, SYSPLLREFCLK_L1C_0, TXCLK19, Data_Ctrl19					BH118			
5B		IO	HVIO_5B_20, TXCLK20, Data_Ctrl20					BF120			
6A		IO	HVIO_6A_1, SYSPLLREFCLK_R4A_0, TXCLK1, Data_Ctrl1					BU28			
6A		IO	HVIO_6A_2, SYSPLLREFCLK_R4A_1, TXCLK2, Data_Ctrl2					BP31			
6A		IO	HVIO_6A_3, SYSPLLREFCLK_R4B_0, TXCLK3, Data_Ctrl3					BR28			
6A		IO	HVIO_6A_4, SYSPLLREFCLK_R4B_1, TXCLK4, Data_Ctrl4					BR31			
6A		IO	HVIO_6A_5, PIN_PERST_N_R4A_0, TXCLK5, Data_Ctrl5					BU31			
6A		IO	HVIO_6A_6, PIN_PERST_N_R4B_0, TXCLK6, Data_Ctrl6					BM28			
6A		IO	HVIO_6A_7, PIN_PERST_N_R4C_0, TXCLK7, Data_Ctrl7					BW28			
6A		IO	HVIO_6A_8, TXCLK8, Data_Ctrl8					BM31			
6A		IO	HVIO_6A_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					BK31			
6A		IO	HVIO_6A_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					BP22			
6A		IO	HVIO_6A_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					BK28			
6A		IO	HVIO_6A_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					BR22			
6A		IO	HVIO_6A_13, TXCLK13, Data_Ctrl13					CH12			
6A		IO	HVIO_6A_14, TXCLK14, Data_Ctrl14					BU22			
6A		IO	HVIO_6A_15, TXCLK15, Data_Ctrl15					BW19			
6A		IO	HVIO_6A_16, TXCLK16, Data_Ctrl16					BR28			
6A		IO	HVIO_6A_17, TXCLK17, Data_Ctrl17					BM22			
6A		IO	HVIO_6A_18, TXCLK18, Data_Ctrl18					CF12			
6A		IO	HVIO_6A_19, SYSPLLREFCLK_R4C_0, TXCLK19, Data_Ctrl19					BK19			
6A		IO	HVIO_6A_20, TXCLK20, Data_Ctrl20					CF9			
6B		IO	HVIO_6B_1, TXCLK1, Data_Ctrl1					BF21			
6B		IO	HVIO_6B_2, TXCLK2, Data_Ctrl2					BE21			
6B		IO	HVIO_6B_3, TXCLK3, Data_Ctrl3					BE43			
6B		IO	HVIO_6B_4, TXCLK4, Data_Ctrl4					BF40			
6B		IO	HVIO_6B_5, PIN_PERST_N_R4A_1, TXCLK5, Data_Ctrl5					BE29			
6B		IO	HVIO_6B_6, PIN_PERST_N_R4B_1, TXCLK6, Data_Ctrl6					BE25			
6B		IO	HVIO_6B_7, PIN_PERST_N_R4C_1, TXCLK7, Data_Ctrl7					BF32			
6B		IO	HVIO_6B_8, TXCLK8, Data_Ctrl8					BF36			
6B		IO	HVIO_6B_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					BF29			
6B		IO	HVIO_6B_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					BF25			
6B		IO	HVIO_6B_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					BF16			
6B		IO	HVIO_6B_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					BH19			
6B		IO	HVIO_6B_13, TXCLK13, Data_Ctrl13					BK22			
6B		IO	HVIO_6B_14, TXCLK14, Data_Ctrl14					BM19			
6B		IO	HVIO_6B_15, TXCLK15, Data_Ctrl15					BU19			
6B		IO	HVIO_6B_16, TXCLK16, Data_Ctrl16					BR19			
6B		IO	HVIO_6B_17, TXCLK17, Data_Ctrl17					CK2			
6B		IO	HVIO_6B_18, TXCLK18, Data_Ctrl18					CI2			
6B		IO	HVIO_6B_19, TXCLK19, Data_Ctrl19					CK4			
6B		IO	HVIO_6B_20, TXCLK20, Data_Ctrl20					CH4			
6C		IO	HVIO_6C_1, TXCLK1, Data_Ctrl1					F27			
6C		IO	HVIO_6C_2, TXCLK2, Data_Ctrl2					F24			
6C		IO	HVIO_6C_3, TXCLK3, Data_Ctrl3					H27			
6C		IO	HVIO_6C_4, TXCLK4, Data_Ctrl4					D24			
6C		IO	HVIO_6C_5, TXCLK5, Data_Ctrl5					H18			
6C		IO	HVIO_6C_6, TXCLK6, Data_Ctrl6					D15			
6C		IO	HVIO_6C_7, TXCLK7, Data_Ctrl7					F18			
6C		IO	HVIO_6C_8, TXCLK8, Data_Ctrl8					F15			
6C		IO	HVIO_6C_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					D8			
6C		IO	HVIO_6C_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					K8			
6C		IO	HVIO_6C_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					F8			
6C		IO	HVIO_6C_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					H8			
6C		IO	HVIO_6C_13, TXCLK13, Data_Ctrl13					C2			
6C		IO	HVIO_6C_14, TXCLK14, Data_Ctrl14					D4			
6C		IO	HVIO_6C_15, TXCLK15, Data_Ctrl15					F4			
6C		IO	HVIO_6C_16, TXCLK16, Data_Ctrl16					K4			
6C		IO	HVIO_6C_17, TXCLK17, Data_Ctrl17					G2			
6C		IO	HVIO_6C_18, TXCLK18, Data_Ctrl18					J2			
6C		IO	HVIO_6C_19, TXCLK19, Data_Ctrl19					J1			
6C		IO	HVIO_6C_20, TXCLK20, Data_Ctrl20					G1			
6D		IO	HVIO_6D_1, TXCLK1, Data_Ctrl1					A8			
6D		IO	HVIO_6D_2, TXCLK2, Data_Ctrl2					B4			
6D		IO	HVIO_6D_3, TXCLK3, Data_Ctrl3					A11			
6D		IO	HVIO_6D_4, TXCLK4, Data_Ctrl4					B11			
6D		IO	HVIO_6D_5, TXCLK5, Data_Ctrl5					B14			
6D		IO	HVIO_6D_6, TXCLK6, Data_Ctrl6					A14			
6D		IO	HVIO_6D_7, TXCLK7, Data_Ctrl7					A20			
6D		IO	HVIO_6D_8, TXCLK8, Data_Ctrl8					A17			
6D		IO	HVIO_6D_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					A23			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
6D		IO	HVIO_6D_10.PLLREFCLK2_TXCLK10,RXCLK2_Data_Ctrl10					B20			
6D		IO	HVIO_6D_11.SOURCE_SYNC_CLK1,TXCLK11,RXCLK3_Data_Ctrl11					B23			
6D		IO	HVIO_6D_12.SOURCE_SYNC_CLK2,TXCLK12,RXCLK4_Data_Ctrl12					B26			
6D		IO	HVIO_6D_13.TXCLK13_Data_Ctrl13					B30			
6D		IO	HVIO_6D_14.TXCLK14_Data_Ctrl14					A30			
6D		IO	HVIO_6D_15.TXCLK15_Data_Ctrl15					A35			
6D		IO	HVIO_6D_16.TXCLK16_Data_Ctrl16					A33			
6D		IO	HVIO_6D_17.TXCLK17_Data_Ctrl17					A39			
6D		IO	HVIO_6D_18.TXCLK18_Data_Ctrl18					B35			
6D		IO	HVIO_6D_19.TXCLK19_Data_Ctrl19					D34			
6D		IO	HVIO_6D_20.TXCLK20_Data_Ctrl20					B39			
1A		GTSL1A_RX_CH0p						BV135			
1A		GTSL1A_RX_CH0n						BV133			
1A		GTSL1A_RX_CH1p						BN135			
1A		GTSL1A_RX_CH1n						BN133			
1A		GTSL1A_RX_CH2p						BU135			
1A		GTSL1A_RX_CH2n						BU133			
1A		GTSL1A_RX_CH3p						BF135			
1A		GTSL1A_RX_CH3n						BF133			
1A		GTSL1A_TX_CH0p						BY129			
1A		GTSL1A_TX_CH0n						BY126			
1A		GTSL1A_TX_CH1p						BT129			
1A		GTSL1A_TX_CH1n						BT126			
1A		GTSL1A_TX_CH2p						BL129			
1A		GTSL1A_TX_CH2n						BL126			
1A		GTSL1A_TX_CH3p						BG129			
1A		GTSL1A_TX_CH3n						BG126			
1A		REFCLK_GTSL1A_CH1p						BB120			
1A		REFCLK_GTSL1A_CH1n						BB115			
1A		REFCLK_GTSL1A_RX_P						BC111			
1A		REFCLK_GTSL1A_RX_N						BC107			
1B		GTSL1B_RX_CH0p						BD135			
1B		GTSL1B_RX_CH0n						BD133			
1B		GTSL1B_RX_CH1p						BB135			
1B		GTSL1B_RX_CH1n						BB133			
1B		GTSL1B_RX_CH2p						AY135			
1B		GTSL1B_RX_CH2n						AY133			
1B		GTSL1B_RX_CH3p						AV135			
1B		GTSL1B_RX_CH3n						AV133			
1B		GTSL1B_TX_CH0p						BE129			
1B		GTSL1B_TX_CH0n						BE126			
1B		GTSL1B_TX_CH1p						BC129			
1B		GTSL1B_TX_CH1n						BC126			
1B		GTSL1B_TX_CH2p						BA129			
1B		GTSL1B_TX_CH2n						BA126			
1B		GTSL1B_TX_CH3p						AW129			
1B		GTSL1B_TX_CH3n						AW126			
1B		REFCLK_GTSL1B_CH1p						AV120			
1B		REFCLK_GTSL1B_CH1n						AV115			
1B		REFCLK_GTSL1B_RX_P						AY120			
1B		REFCLK_GTSL1B_RX_N						AY115			
1C		GTSL1C_RX_CH0p						AT135			
1C		GTSL1C_RX_CH0n						AT133			
1C		GTSL1C_RX_CH1p						AP135			
1C		GTSL1C_RX_CH1n						AP133			
1C		GTSL1C_RX_CH2p						AM135			
1C		GTSL1C_RX_CH2n						AM133			
1C		GTSL1C_RX_CH3p						AK135			
1C		GTSL1C_RX_CH3n						AK133			
1C		GTSL1C_TX_CH0p						AU129			
1C		GTSL1C_TX_CH0n						AU126			
1C		GTSL1C_TX_CH1p						AR129			
1C		GTSL1C_TX_CH1n						AR126			
1C		GTSL1C_TX_CH2p						AN129			
1C		GTSL1C_TX_CH2n						AN126			
1C		GTSL1C_TX_CH3p						AL129			
1C		GTSL1C_TX_CH3n						AL126			
1C		REFCLK_GTSL1C_CH1p						AP120			
1C		REFCLK_GTSL1C_CH1n						AP115			
1C		CDRCLKOUT_GTSL1C_CH2p						AN111			
1C		CDRCLKOUT_GTSL1C_CH2n						AN107			
1C		REFCLK_GTSL1C_RX_P						AT120			
1C		REFCLK_GTSL1C_RX_N						AT115			
4A		GTSR4A_RX_CH0p						CB1			
4A		GTSR4A_RX_CH0n						CB3			
4A		GTSR4A_RX_CH1p						BV1			
4A		GTSR4A_RX_CH1n						BV3			
4A		GTSR4A_RX_CH2p						BN1			
4A		GTSR4A_RX_CH2n						BN3			
4A		GTSR4A_RX_CH3p						BU1			
4A		GTSR4A_RX_CH3n						BU3			
4A		GTSR4A_TX_CH0p						BV7			

altera

Pin Information for Agilex™ 5 A5ED052A Device

Version: 2025-06-06

B32A Status: Final

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
4A		GTSR4A_TX_CH0n						BV10			
4A		GTSR4A_TX_CH1p						BT7			
4A		GTSR4A_TX_CH1n						BT10			
4A		GTSR4A_TX_CH2p						BL7			
4A		GTSR4A_TX_CH2n						BL10			
4A		GTSR4A_TX_CH3p						BG7			
4A		GTSR4A_TX_CH3n						BG10			
4A		REFCLK_GTSR4A_CH1p						BB16			
4A		REFCLK_GTSR4A_CH1n						BB21			
4A		REFCLK_GTSR4A_RX_P						BC29			
4A		REFCLK_GTSR4A_RX_N						BC25			
4B		GTSR4B_RX_CH0p						BF1			
4B		GTSR4B_RX_CH0n						BF3			
4B		GTSR4B_RX_CH1p						BD1			
4B		GTSR4B_RX_CH1n						BD3			
4B		GTSR4B_RX_CH2p						BB1			
4B		GTSR4B_RX_CH2n						BB3			
4B		GTSR4B_RX_CH3p						AY1			
4B		GTSR4B_RX_CH3n						AY3			
4B		GTSR4B_TX_CH0p						BE7			
4B		GTSR4B_TX_CH0n						BE10			
4B		GTSR4B_TX_CH1p						BC7			
4B		GTSR4B_TX_CH1n						BC10			
4B		GTSR4B_TX_CH2p						BA7			
4B		GTSR4B_TX_CH2n						BA10			
4B		GTSR4B_TX_CH3p						AW7			
4B		GTSR4B_TX_CH3n						AW10			
4B		REFCLK_GTSR4B_CH1p						AV16			
4B		REFCLK_GTSR4B_CH1n						AV21			
4B		CDRCLKOUT_GTSR4B_CH2p						AN25			
4B		CDRCLKOUT_GTSR4B_CH2n						AN29			
4B		REFCLK_GTSR4B_RX_P						AY16			
4B		REFCLK_GTSR4B_RX_N						AY21			
4C		GTSR4C_RX_CH0p						AV1			
4C		GTSR4C_RX_CH0n						AV3			
4C		GTSR4C_RX_CH1p						AT1			
4C		GTSR4C_RX_CH1n						AT3			
4C		GTSR4C_RX_CH2p						AP1			
4C		GTSR4C_RX_CH2n						AP3			
4C		GTSR4C_RX_CH3p						AM1			
4C		GTSR4C_RX_CH3n						AM3			
4C		GTSR4C_TX_CH0p						AU7			
4C		GTSR4C_TX_CH0n						AU10			
4C		GTSR4C_TX_CH1p						AR7			
4C		GTSR4C_TX_CH1n						AR10			
4C		GTSR4C_TX_CH2p						AN7			
4C		GTSR4C_TX_CH2n						AN10			
4C		GTSR4C_TX_CH3p						AL7			
4C		GTSR4C_TX_CH3n						AL10			
4C		REFCLK_GTSR4C_CH1p						AP16			
4C		REFCLK_GTSR4C_CH1n						AP21			
4C		REFCLK_GTSR4C_RX_P						AT16			
4C		REFCLK_GTSR4C_RX_N						AT21			
		GND						CK110			
		GND						CH121			
		GND						CH118			
		GND						V95			
		GND						V84			
		GND						V74			
		GND						V65			
		GND						V55			
		GND						V44			
		GND						V4			
		GND						V34			
		GND						V24			
		GND						V15			
		GND						V132			
		GND						V124			
		GND						V114			
		GND						V105			
		GND						R135			
		GND						R1			
		GND						P98			
		GND						P87			
		GND						P8			
		GND						P77			
		GND						P67			
		GND						P58			
		GND						P47			
		GND						P37			
		GND						P27			
		GND						P18			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						P127			
		GND						P117			
		GND						P108			
		GND						L2			
		GND						L134			
		GND						H95			
		GND						H84			
		GND						H74			
		GND						H65			
		GND						H55			
		GND						H44			
		GND						H4			
		GND						H34			
		GND						H24			
		GND						H15			
		GND						H132			
		GND						H124			
		GND						H114			
		GND						H105			
		GND						E2			
		GND						E134			
		GND						E1			
		GND						D98			
		GND						D87			
		GND						D77			
		GND						D67			
		GND						D58			
		GND						D47			
		GND						D37			
		GND						D27			
		GND						D18			
		GND						D127			
		GND						D117			
		GND						D108			
		GND						CL94			
		GND						CL80			
		GND						CL63			
		GND						CL48			
		GND						CL4			
		GND						CL33			
		GND						CL2			
		GND						CL134			
		GND						CL132			
		GND						CL122			
		GND						CL110			
		GND						CK91			
		GND						CK82			
		GND						CK70			
		GND						CK60			
		GND						CK6			
		GND						CK51			
		GND						CK42			
		GND						CK23			
		GND						CK14			
		GND						CK135			
		GND						CK132			
		GND						CK130			
		GND						CK122			
		GND						CK106			
		GND						CK1			
		GND						CJ135			
		GND						CJ134			
		GND						CJ1			
		GND						CH9			
		GND						CH28			
		GND						CH19			
		GND						CH112			
		GND						CH102			
		GND						CE5			
		GND						CE3			
		GND						CE1			
		GND						CC99			
		GND						CC89			
		GND						CC78			
		GND						CC69			
		GND						CC59			
		GND						CC49			
		GND						CC38			
		GND						CC118			
		GND						CC112			
		GND						CC108			
		GND						CB7			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						CB5			
		GND						CB131			
		GND						CB13			
		GND						CB129			
		GND						CB126			
		GND						CB123			
		GND						CB10			
		GND						CA28			
		GND						CA19			
		GND						C135			
		GND						C134			
		GND						C1			
		GND						BY5			
		GND						BY3			
		GND						BY135			
		GND						BY133			
		GND						BY131			
		GND						BY13			
		GND						BY123			
		GND						BY1			
		GND						BW92			
		GND						BW81			
		GND						BW71			
		GND						BW62			
		GND						BW52			
		GND						BW41			
		GND						BW31			
		GND						BW22			
		GND						BV7			
		GND						BV5			
		GND						BV131			
		GND						BV13			
		GND						BV129			
		GND						BV126			
		GND						BV123			
		GND						BV10			
		GND						BU112			
		GND						BU102			
		GND						BT5			
		GND						BT3			
		GND						BT135			
		GND						BT133			
		GND						BT131			
		GND						BT13			
		GND						BT123			
		GND						BT1			
		GND						BP99			
		GND						BP89			
		GND						BP78			
		GND						BP69			
		GND						BP59			
		GND						BP49			
		GND						BP38			
		GND						BP28			
		GND						BP19			
		GND						BP118			
		GND						BP109			
		GND						BN7			
		GND						BN5			
		GND						BN131			
		GND						BN13			
		GND						BN129			
		GND						BN126			
		GND						BN123			
		GND						BN10			
		GND						BL5			
		GND						BL3			
		GND						BL135			
		GND						BL133			
		GND						BL131			
		GND						BL13			
		GND						BL123			
		GND						BL1			
		GND						BK92			
		GND						BK81			
		GND						BK71			
		GND						BK62			
		GND						BK52			
		GND						BK41			
		GND						BJ7			
		GND						BJ5			
		GND						BJ131			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						BJ13			
		GND						BJ29			
		GND						BJ26			
		GND						BJ23			
		GND						BJ10			
		GND						BH31			
		GND						BH22			
		GND						BH112			
		GND						BH102			
		GND						BG5			
		GND						BG3			
		GND						BG135			
		GND						BG133			
		GND						BG131			
		GND						BG13			
		GND						BG123			
		GND						BG1			
		GND						BF96			
		GND						BF79			
		GND						BF7			
		GND						BF61			
		GND						BF5			
		GND						BF43			
		GND						BF131			
		GND						BF13			
		GND						BF129			
		GND						BF126			
		GND						BF123			
		GND						BF10			
		GND						BE90			
		GND						BE72			
		GND						BE53			
		GND						BE5			
		GND						BE40			
		GND						BE36			
		GND						BE32			
		GND						BE3			
		GND						BE16			
		GND						BE135			
		GND						BE133			
		GND						BE131			
		GND						BE13			
		GND						BE123			
		GND						BE120			
		GND						BE104			
		GND						BE1			
		GND						BD93			
		GND						BD79			
		GND						BD7			
		GND						BD68			
		GND						BD67			
		GND						BD6			
		GND						BD46			
		GND						BD40			
		GND						BD32			
		GND						BD29			
		GND						BD25			
		GND						BD21			
		GND						BD131			
		GND						BD13			
		GND						BD129			
		GND						BD126			
		GND						BD123			
		GND						BD115			
		GND						BD111			
		GND						BD107			
		GND						BD104			
		GND						BD10			
		GND						BC93			
		GND						BC83			
		GND						BC72			
		GND						BC61			
		GND						BC50			
		GND						BC5			
		GND						BC32			
		GND						BC3			
		GND						BC21			
		GND						BC16			
		GND						BC136			
		GND						BC133			
		GND						BC131			
		GND						BC13			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						BC123			
		GND						BC120			
		GND						BC115			
		GND						BC104			
		GND						BC1			
		GND						BB86			
		GND						BB75			
		GND						BB7			
		GND						BB64			
		GND						BB53			
		GND						BB5			
		GND						BB36			
		GND						BB32			
		GND						BB29			
		GND						BB25			
		GND						BB131			
		GND						BB13			
		GND						BB129			
		GND						BB126			
		GND						BB123			
		GND						BB111			
		GND						BB107			
		GND						BB104			
		GND						BB100			
		GND						BB10			
		GND						BA96			
		GND						BA93			
		GND						BA90			
		GND						BA79			
		GND						BA68			
		GND						BA57			
		GND						BA5			
		GND						BA46			
		GND						BA43			
		GND						BA40			
		GND						BA36			
		GND						BA3			
		GND						BA25			
		GND						BA21			
		GND						BA16			
		GND						BA135			
		GND						BA133			
		GND						BA131			
		GND						BA13			
		GND						BA123			
		GND						BA120			
		GND						BA115			
		GND						BA111			
		GND						BA100			
		GND						BA1			
		GND						BB4			
		GND						BB0			
		GND						B6			
		GND						BB3			
		GND						B6			
		GND						BB48			
		GND						B33			
		GND						B2			
		GND						B17			
		GND						B135			
		GND						B132			
		GND						B125			
		GND						B110			
		GND						AY83			
		GND						AY72			
		GND						AY7			
		GND						AY61			
		GND						AY5			
		GND						AY32			
		GND						AY131			
		GND						AY13			
		GND						AY129			
		GND						AY126			
		GND						AY123			
		GND						AY104			
		GND						AY10			
		GND						AW96			
		GND						AW90			
		GND						AW86			
		GND						AW75			
		GND						AW64			
		GND						AW53			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						AW5			
		GND						AW46			
		GND						AW40			
		GND						AW3			
		GND						AW25			
		GND						AW21			
		GND						AW16			
		GND						AW135			
		GND						AW133			
		GND						AW131			
		GND						AW13			
		GND						AW123			
		GND						AW120			
		GND						AW115			
		GND						AW111			
		GND						AW1			
		GND						AV96			
		GND						AV79			
		GND						AV7			
		GND						AV68			
		GND						AV57			
		GND						AV5			
		GND						AV40			
		GND						AV32			
		GND						AV29			
		GND						AV25			
		GND						AV131			
		GND						AV13			
		GND						AV129			
		GND						AV126			
		GND						AV123			
		GND						AV111			
		GND						AV107			
		GND						AV104			
		GND						AV10			
		GND						AU90			
		GND						AU83			
		GND						AU61			
		GND						AU5			
		GND						AU46			
		GND						AU3			
		GND						AU21			
		GND						AU16			
		GND						AU135			
		GND						AU133			
		GND						AU131			
		GND						AU13			
		GND						AU123			
		GND						AU120			
		GND						AU115			
		GND						AU1			
		GND						AT96			
		GND						AT86			
		GND						AT75			
		GND						AT7			
		GND						AT64			
		GND						AT53			
		GND						AT5			
		GND						AT40			
		GND						AT36			
		GND						AT32			
		GND						AT29			
		GND						AT131			
		GND						AT13			
		GND						AT129			
		GND						AT126			
		GND						AT123			
		GND						AT107			
		GND						AT104			
		GND						AT100			
		GND						AT10			
		GND						AR96			
		GND						AR90			
		GND						AR79			
		GND						AR68			
		GND						AR57			
		GND						AR5			
		GND						AR46			
		GND						AR40			
		GND						AR3			
		GND						AR25			
		GND						AR21			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						AR16			
		GND						AR135			
		GND						AR133			
		GND						AR131			
		GND						AR13			
		GND						AR123			
		GND						AR120			
		GND						AR115			
		GND						AR111			
		GND						AR1			
		GND						AP83			
		GND						AP72			
		GND						AP7			
		GND						AP61			
		GND						AP5			
		GND						AP32			
		GND						AP131			
		GND						AP13			
		GND						AP129			
		GND						AP126			
		GND						AP123			
		GND						AP104			
		GND						AP10			
		GND						AN96			
		GND						AN93			
		GND						AN86			
		GND						AN75			
		GND						AN64			
		GND						AN53			
		GND						AN5			
		GND						AN46			
		GND						AN43			
		GND						AN40			
		GND						AN36			
		GND						AN32			
		GND						AN3			
		GND						AN21			
		GND						AN16			
		GND						AN135			
		GND						AN133			
		GND						AN131			
		GND						AN13			
		GND						AN123			
		GND						AN120			
		GND						AN115			
		GND						AN104			
		GND						AN100			
		GND						AN1			
		GND						AM90			
		GND						AM79			
		GND						AM7			
		GND						AM68			
		GND						AM57			
		GND						AM5			
		GND						AM32			
		GND						AM29			
		GND						AM25			
		GND						AM21			
		GND						AM131			
		GND						AM13			
		GND						AM129			
		GND						AM126			
		GND						AM123			
		GND						AM115			
		GND						AM111			
		GND						AM107			
		GND						AM104			
		GND						AM10			
		GND						AL93			
		GND						AL83			
		GND						AL72			
		GND						AL61			
		GND						AL50			
		GND						AL5			
		GND						AL36			
		GND						AL3			
		GND						AL21			
		GND						AL135			
		GND						AL133			
		GND						AL131			
		GND						AL13			
		GND						AL123			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						AL111			
		GND						AL1			
		GND						AK96			
		GND						AK86			
		GND						AK75			
		GND						AK7			
		GND						AK64			
		GND						AK53			
		GND						AK5			
		GND						AK43			
		GND						AK36			
		GND						AK29			
		GND						AK131			
		GND						AK13			
		GND						AK129			
		GND						AK126			
		GND						AK123			
		GND						AK100			
		GND						AK10			
		GND						AG96			
		GND						AG86			
		GND						AG68			
		GND						AG50			
		GND						AG32			
		GND						AG25			
		GND						AG16			
		GND						AG135			
		GND						AG133			
		GND						AG131			
		GND						AG126			
		GND						AG107			
		GND						AF1			
		GND						AE8			
		GND						AC93			
		GND						AC75			
		GND						AC57			
		GND						AC40			
		GND						AA2			
		GND						AA134			
		GND						A88			
		GND						A73			
		GND						A6			
		GND						A56			
		GND						A42			
		GND						A4			
		GND						A26			
		GND						A134			
		GND						A132			
		GND						A119			
		GND						A103			
		GNDSENSE						AU72			
		VCC						BA61			
		VCC						AV75			
		VCC						AV68			
		VCC						AV64			
		VCC						AV57			
		VCC						AV53			
		VCC						AW79			
		VCC						AW72			
		VCC						AW68			
		VCC						AW61			
		VCC						AW57			
		VCC						AV86			
		VCC						AV83			
		VCC						AV75			
		VCC						AV64			
		VCC						AV61			
		VCC						AV53			
		VCC						AU79			
		VCC						AU75			
		VCC						AU68			
		VCC						AU64			
		VCC						AU57			
		VCC						AU53			
		VCC						AT79			
		VCC						AT72			
		VCC						AT68			
		VCC						AT61			
		VCC						AT57			
		VCC						AR83			
		VCC						AR75			
		VCC						AR72			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		VCC						AP64			
		VCC						AP61			
		VCC						AP53			
		VCC						AP86			
		VCC						AP79			
		VCC						AP75			
		VCC						AP68			
		VCC						AP64			
		VCC						AN79			
		VCC						AN72			
		VCCPT						BA75			
		VCCPT						BA72			
		VCCPT						AP57			
		VCCPT						AN61			
		VCCPT_HVIO						BC90			
		VCCPT_HVIO						BB90			
		VCCPT_HVIO						BB46			
		VCCPT_HVIO						BA50			
		VCCPT_HVIO						AL46			
		VCCPT_HVIO						AK46			
		DNU						CL101			
		DNU						CK101			
		DNU						CK103			
		DNU						CL106			
		DNU						AM120			
		DNU						AL115			
		DNU						BD64			
		DNU						BD53			
		DNU						AL68			
		DNU						AL57			
		DNU						BD120			
		DNU						BC40			
		DNU						AL43			
		DNU						BA104			
		DNU						BA32			
		DNU						AR104			
		DNU						AR32			
		DNU						BD100			
		DNU						BD36			
		DNU						AM40			
		DNU						BC100			
		DNU						BC36			
		DNU						AM36			
		VCCBAT						BD86			
		VCCIO_HPS						AL90			
		VCCIO_HPS						AK90			
		VCCIO_HVIO_5A						BD96			
		VCCIO_HVIO_5A						BC96			
		VCCIO_HVIO_5B						BB96			
		VCCIO_HVIO_5B						BB93			
		VCCIO_HVIO_6A						BB43			
		VCCIO_HVIO_6A						BB40			
		VCCIO_HVIO_6B						BD43			
		VCCIO_HVIO_6B						BC43			
		VCCIO_HVIO_6C						AL40			
		VCCIO_HVIO_6C						AK40			
		VCCIO_HVIO_6D						AG46			
		VCCIO_HVIO_6D						AG43			
		VCCIO_PIO_2A_B						BD72			
		VCCIO_PIO_2A_B						BC68			
		VCCIO_PIO_2A_T						BD61			
		VCCIO_PIO_2A_T						BC64			
		VCCIO_PIO_2B_B						BC57			
		VCCIO_PIO_2B_B						BC53			
		VCCIO_PIO_2B_T						BD50			
		VCCIO_PIO_2B_T						BC46			
		VCCIO_PIO_3A_B						AL75			
		VCCIO_PIO_3A_B						AK72			
		VCCIO_PIO_3A_T						AL64			
		VCCIO_PIO_3A_T						AK68			
		VCCIO_PIO_3B_B						AK61			
		VCCIO_PIO_3B_B						AK57			
		VCCIO_PIO_3B_T						AL53			
		VCCIO_PIO_3B_T						AK50			
		VCCIO_PIO_SDM						AW83			
		VCCIO_SDM						BD75			
		NC						AL104			
		NC						AL107			
		NC						AM43			
		NC						AM46			
		NC						AM50			
		NC						AN50			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		NC						AI50			
		NC						AI50			
		NC						AI50			
		NC						AI50			
		NC						AP50			
		NC						AP53			
		NC						AR50			
		NC						AT50			
		NC						AM16			
		NC						BD16			
		NC						AG29			
		NC						AL29			
		NC						AK21			
		NC						AG21			
		NC						AK25			
		NC						AL25			
		NC						Y27			
		NC						AB27			
		NC						AB24			
		NC						Y24			
		NC						T27			
		NC						V27			
		NC						P24			
		NC						T24			
		NC						K27			
		NC						M27			
		NC						M24			
		NC						K24			
		NC						AK16			
		NC						AL16			
		NC						P4			
		NC						M4			
		NC						N1			
		NC						L1			
		NC						T4			
		NC						N2			
		NC						U2			
		NC						R2			
		NC						U1			
		NC						W1			
		NC						W2			
		NC						Y4			
		NC						AE4			
		NC						AB4			
		NC						AF2			
		NC						AD2			
		NC						AJ2			
		NC						AA1			
		NC						AJ1			
		NC						AD1			
		NC						F37			
		NC						F34			
		NC						H37			
		NC						K34			
		NC						P34			
		NC						K37			
		NC						M34			
		NC						M37			
		NC						Y37			
		NC						Y34			
		NC						AC36			
		NC						V37			
		NC						T34			
		NC						AC46			
		NC						T37			
		NC						AL32			
		NC						AG40			
		NC						AG36			
		NC						AC43			
		NC						AK32			
		NC						K15			
		NC						M15			
		NC						V8			
		NC						T8			
		NC						M8			
		NC						AB8			
		NC						Y8			
		NC						AB4			
		NC						K18			
		NC						Y15			
		NC						M18			
		NC						V18			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		NC						T16			
		NC						AN8			
		NC						T15			
		NC						P15			
		NC						AG13			
		NC						Y18			
		NC						AB15			
		NC						AB18			
		VCCRCORE						AU86			
		VCCRCORE						AT83			
		VCCRCORE						AR86			
		VCCADC						BD83			
		VCCEHT_GTSL1A						AY111			
		VCCEHT_GTSL1A						AY107			
		VCCEHT_GTSL1B						AU111			
		VCCEHT_GTSL1B						AT111			
		VCCEHT_GTSL1C						AP111			
		VCCEHT_GTSL1C						AP107			
		VCCEHT_GTSR4A						AY29			
		VCCEHT_GTSR4A						AY25			
		VCCEHT_GTSR4B						AU25			
		VCCEHT_GTSR4B						AT25			
		VCCEHT_GTSR4C						AP29			
		VCCEHT_GTSR4C						AP25			
		VCCERT_GTSL1A						AY96			
		VCCERT_GTSL1A						AY100			
		VCCERT_GTSL1A						AW100			
		VCCERT_GTSL1B						AU100			
		VCCERT_GTSL1B						AU96			
		VCCERT_GTSL1B						AU100			
		VCCERT_GTSL1C						AR100			
		VCCERT_GTSL1C						AP96			
		VCCERT_GTSL1C						AP100			
		VCCERT_GTSR4A						AY40			
		VCCERT_GTSR4A						AY36			
		VCCERT_GTSR4A						AW36			
		VCCERT_GTSR4B						AV36			
		VCCERT_GTSR4B						AU40			
		VCCERT_GTSR4B						AU36			
		VCCERT_GTSR4C						AR36			
		VCCERT_GTSR4C						AP40			
		VCCERT_GTSR4C						AP36			
		VCCFUSEWR_SDM						BC86			
		VCCH_SDM						BA86			
		VCCLSENSE						AV72			
		VCCL_ADC_SDM						AY86			
		VCCL_HPS						AN83			
		VCCL_HPS						AM83			
		VCCL_HPS						AL86			
		VCCL_HPS						AK83			
		VCCL_HPS_CORE0_CORE1						AL79			
		VCCL_HPS_CORE0_CORE1						AK79			
		VCCL_HPS_CORE2						AL96			
		VCCL_HPS_CORE2						AK93			
		VCCL_HPS_CORE3						AM100			
		VCCL_HPS_CORE3						AL100			
		VCCL_SDM						BB83			
		VCCL_SDM						BB79			
		VCCL_SDM						BA83			
		VCCL_SDM						AY79			
		VCCP						BB72			
		VCCP						BB68			
		VCCP						BB61			
		VCCP						BB57			
		VCCP						BB50			
		VCCP						BA64			
		VCCP						BA53			
		VCCP						AN68			
		VCCP						AN57			
		VCCP						AM75			
		VCCP						AM72			
		VCCP						AM64			
		VCCP						AM61			
		VCCP						AM53			
		VCCPLL1_HPS						AM93			
		VCCPLL2_HPS						AN90			
		VCCPLLDIG1_HPS						AM96			
		VCCPLLDIG2_HPS						AM96			
		VCCPLLDIG2_HPS						BC75			
		VCCPLLDIG2_SDM						BC79			
		VCCPLL_SDM						AY93			
		VCC_HSSI_L1						AY90			
		VCC_HSSI_L1									

altera Pin Information for Agilex™ 5 A5ED052A Device Version: 2025-06-06 B32A Status: Final											
Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B32A	DQS for X4	DQS for X8/X9	DQS for X16/X18
		VCC_HSSI_L1						AW93			
		VCC_HSSI_L1						AV93			
		VCC_HSSI_L1						AU93			
		VCC_HSSI_L1						AT93			
		VCC_HSSI_L1						AP93			
		VCC_HSSI_L1						AR93			
		VCC_HSSI_L1						AP90			
		VCC_HSSI_R4						AY46			
		VCC_HSSI_R4						AY43			
		VCC_HSSI_R4						AW43			
		VCC_HSSI_R4						AV46			
		VCC_HSSI_R4						AV43			
		VCC_HSSI_R4						AU43			
		VCC_HSSI_R4						AT46			
		VCC_HSSI_R4						AT43			
		VCC_HSSI_R4						AR43			
		VCC_HSSI_R4						AP46			
		VCC_HSSI_R4						AP43			
		VCC_IO_SDM						BD90			
		APROBE2_GTSL1C						AW104			
		APROBE2_GTSR4C						AW32			
		APROBE_GTSL1A_CH3						BA107			
		APROBE_GTSL1C_CH2						AW107			
		APROBE_GTSL1C_CH3						AR107			
		APROBE_GTSR4A_CH3						BA29			
		APROBE_GTSR4B_CH3						AW29			
		APROBE_GTSR4C_CH3						AR29			
		RCOMP_GTSL1B_N						AU104			
		RCOMP_GTSL1B_P						AU107			
		RCOMP_GTSR4B_N						AU32			
		RCOMP_GTSR4B_P						AU29			

Date	Version	Revision History for B23A Package
August 2024	2024-08-19	Initial release.
January 2025	2025-01-13	Added I3C0_SDA_PULLUP_EN and I3C1_SDA_PULLUP_EN signal.
June 2025	2025-06-06	Updated soft CDR support column: -Added CDR support on pin index 0, 1, 12, 13, 24, 25, 36, and 37. -Removed CDR support on pin index 10, 11, 22, 23, 34, 35, 46, and 47.

Date	Version	Revision History for B32A Package
August 2024	2024-08-19	Initial release.
January 2025	2025-01-13	Added I3C0_SDA_PULLUP_EN and I3C1_SDA_PULLUP_EN signal.
June 2025	2025-06-06	Updated soft CDR support column: -Added CDR support on pin index 0, 1, 12, 13, 24, 25, 36, and 37. -Removed CDR support on pin index 10, 11, 22, 23, 34, 35, 46, and 47.

(1) For more information about pin definition and pin connection guidelines, refer to the

[Pin Connection Guidelines: Agilex™ 5 FPGAs and SoCs](#)

(2) IOB for IO pin name/function is equivalent to HSIO.

(3) To obtain a Comma Delimited (CSV) file using Excel - From the active tab, select "Save As" in the FILE menu and then select CSV (*.csv) format and SAVE.

CSV File may be viewed in a TEXT editor or in Excel.